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Mathematical Logic for the Humanities

Ryan P. Dunning

St. Mary's University

Contents

To the Student

In this course, we will discover how to investigate mathematical results, how to decide for ourselves whether a statement is true, and how to formulate proofs which communicate our certainty of a statement's truth. This all starts with foundational language from symbolic logic, which will be our focus in Chapter 1.

It will be to your benefit to use your time to read these notes carefully. Such careful reading consists of (at least) the following techniques.

1. Each new topic typically begins with a brief discussion and a definition of a new term. Read these definitions more than once, so that you are confident of the new term's meaning. In other disciplines, you have some leeway to miss a new term's meaning, yet pick it up later, from context clues. In mathematics, you do not have that luxury. Also, most definitions contain words that were themselves defined previously in the notes, so consider quizzing yourself on *those* definitions and then flipping back through the notes to re-read *those* definitions to see if you were correct; this may very well help you to understand the new term (and to improve your remembrance of the old term).

2. New definitions may or may not be followed immediately by an example that illustrates the new term. This depends on how complicated I deem the new definition to be. In either event, I recommend that you try to create your own examples for newly defined terms. This will not only help you better grasp the new terms in this course, but it's also a good reading technique in any subject.

3. After the introduction of new topics (definitions, examples, brief motivation, etc.), the notes will contain problems for you to complete. By completing these problems, you will (a) confirm your understanding of the new definitions, (b) unify all of your observations about a particular issue toward the goal of discovering concepts that are new to you, and (c) apply your new discoveries to new situations. In order to achieve these grand goals that the problems are meant to help you attain, you will need to give yourself the time and the freedom to be confused. By this, I mean that a problem could make you reconsider something

that you read previously. When that happens, go back and re-read that item, whether it's a definition, an example, or *an earlier problem*. Pay attention to this last item, in particular. If you forget a problem the instant that you've completed it, then you will not be in good shape to complete the next problem. Therefore, I recommend that when you read a new problem for the first time, re-read the previous problems and ask youself, "What do these previous problems tell me about the current problem?"

As a style note, throughout this document, I will indicate the end of a definition, example, problem, etc., by using the $\square$ symbol.

Chapter 1

Logic

Have you ever found yourself re-reading a complicated sentence in a book because you didn't grasp its full meaning the first time? Perhaps you knew all of the words, but something about the way they were assembled obscured the author's intent. While this can be frustrating at times, we know that life would be less interesting if we only spoke and wrote in short, choppy sentences like "The sky is blue" or "I am tired." In order to convey deep observations and to connect ideas together, we all tend to build complicated sentences out of simpler sentences. In this chapter, we will discuss some standard ways to build complicated sentences, and we will investigate how these building techniques affect the intended meaning of the constructed sentence.

1.1 Statements

To begin our investigation of logic, we first define the fundamental objects of our study: *statements*.

Definition 1. *A* statement *is a complete declarative sentence that is true or false, but not both. By a statement's* truth value*, we mean which one of the labels, "true" or "false," is applied to that statement.* □

Definition 1 does not technically say what a "complete declarative sentence" is. For our immediate purposes, interpret that phrase as it would have been described in a grammar course. Definition 1 also does not technically say what the words "true" and "false" mean. Again, for our immediate purposes, interpret those words exactly as your intuition and experience tell you to. Chapter 2 will address the issue of *undefined terms*, and we will revisit Definition 1 in that context, but for now, keep reading, starting with our first example below.

Example 2. *Each of the following sentences is a statement.*

1. *Kim Ng began serving as the general manager of the Miami Marlins in 2020.*
2. *A regulation soccer goal is taller than a regulation ice hockey goal.*
3. *The word "geometry" comes from the Greek for "terrible lizard."* □

Not that it matters significantly to us right now, but the first two statements in Example 2 are true, and the last statement is false. At this early stage in the course, it is more significant that the truth value of each statement is ultimately not a matter of opinion.

Problem 3. *Explain (using a complete sentence for each) why each of the following is not a statement.*

1. *dog food in the cabinet*
2. *When was the meeting canceled?*
3. *Turn off the air conditioner.* □

You have seen in previous math classes the use of a letter to substitute for a number ("Let x equal 5," for example). Throughout the rest of these notes, we will often let a letter name substitute for an entire statement. For example, if we let P be the statement "Hydrogen atoms have one proton," then the letter name P can take the place of that complete sentence for the remainder of the relevant discussion. Such naming of statements acts as a space-saver and, more importantly, as a means to keep track of changes made to one or more statements. Negations, the subject of the next section, are the first of many such changes in this chapter.

1.2 Negations

In this section, we begin to explore the skill of refuting peoples' claims. That is, *negations* will show us how to start a proper argument with someone.

Definition 4. *Let P be a statement. We denote by $\neg P$ (read out loud as "not P") the* negation *of the statement P, which is the statement that denies the truth value of the statement P.* □

Warning: Note that the symbol for negation is $\neg$, which is not a negative sign. □

Since we traditionally treat "true" and "false" to be each other's opposites, we treat the statements P and $\neg P$ to be each other's opposites, regardless of which specific statement P we're discussing.

Example 5. *Let P be the statement "Babe Ruth hit exactly 714 home runs in his career." Then the statement $\neg P$ is "Babe Ruth did not hit exactly 714 home runs in his career." We can see that P and $\neg P$ are indeed opposite statements. It happens that P is a true statement, and its negation $\neg P$ is false.* □

Example 6. *Let Q be the statement "Hank Aaron hit exactly 800 home runs in his career." Then the statement $\neg Q$ is "Hank Aaron did not hit exactly 800 home runs in his career." The statements Q and $\neg Q$ are opposites of each other. Here, Q is a false statement, and $\neg Q$ is a true statement.* □

Problem 7. *Write the negation of each of the following three statements. Each answer provided should simply be a sentence; there is no need to give these statements names, like P or Q.*

1. *Spanish I is a requirement for Spanish II.*
2. *The usher didn't open the door.*
3. *Vermont is north of New York.* □

In order to better understand how to write the negation of more complicated statements, there are some vital relationships between a statement and its negation that need to be addressed. When considering a statement and its negation:

1. They can't both be true.
2. They can't both be false.
3. One is true, and one is false.
4. One's being true indicates that the other is false.
5. One's being false indicates that the other is true.

I often like to discuss negations in terms of having (or at least, starting) an argument with someone. If I make a statement, and if you want to argue with me, then you claim the opposite of what I claim. That is, you claim the *negation* of my original statement. In this case:

1. You and I can't both be right.
2. You and I can't both be wrong.
3. One of us is right, and the other is wrong.
4. My being right would indicate that you are wrong.
5. My being wrong would indicate that you are right.

6. Your being right would indicate that I am wrong.

7. Your being wrong would indicate that I am right.

Said another way, if you're given a statement that describes some particular situation, then that statement's negation should incorporate all other possibilities for that situation.

Problem 8. *Suppose that a homework problem asks you to write the negation of the statement "Maria is wearing shoes that don't have laces." Suppose that a particular student's response is "Maria is wearing shoes that do have laces."*

1. *This student's response is not actually the negation of the given statement. Briefly explain why, by describing a single situation involving Maria and shoes in which the given statement and the student's attempted negation would both be false. [You can address this by saying, "If (blank), then the given statement and the student's attempted negation would both be false."]*

2. *Write a statement that is actually the negation of the given statement.* □

Problem 9. *Suppose that your uncle never tells the truth. He claims, "The high temperature at the airport yesterday was strictly between 30 and 40 degrees."*

1. *What are the possible high temperatures at the airport for that day, knowing that temperatures strictly between 30 and 40 degrees are incorrect?*

2. *Write a negation of your uncle's claim. To do this, use the word "or" an appropriate number of times, so that your response is more substantial than simply adding a word like "not" or a phrase like "anything but" to his claim. [I am asking you to make your response more substantial than the approach illustrated in Example 5.]* □

With Problem 9 in mind, note that negating a claim doesn't necessarily settle an argument. However, negating a claim always identifies the opposing sides of an argument.

Example 10. *Roger claims, "Every U.S. President went to college." Ana wants to argue with Roger about this claim, so she needs to declare her side of the argument. She does this by negating his claim. That negation is "At least one U.S. President did not go to college" because that covers all possibilities beyond the one described by Roger. Now the lines have been drawn; Roger's being right would indicate that Ana is wrong, and Roger's being* *wrong would indicate that Ana is right. Identifying the negation doesn't settle the argument, though, so Roger and Ana both do some research and find that Abraham Lincoln did not go to college. Note that "Abraham Lincoln did not go to college" is not the negation of Roger's initial claim, but it is hard evidence that Roger is wrong and Ana is right.* □

Warning: Students are often tempted to negate the statement "Every U.S. President went to college" by saying that "Not every U.S. President went to college." This is problematic because "Not every U.S. President went to college" has potentially questionable grammar (Do you mean that none did or that some didn't? "Some didn't" is probably what you meant, and would be a lot more clear). Moving forward, when a given statement is complicated enough, we will never write the negation by simply adding the word "not" to that statement. □

Warning: Technically, we can always negate a statement P by saying "It is not the case that..." and pasting on an exact copy of the statement P. However, this is cheap, and it sometimes results in an unclear sentence. Therefore, we will agree to avoid phrasing negations that way. □

Problem 11. *Write the negation of each of the following three statements. Make your answers more substantial than simply adding the word "not" to the given statement.*

1. *Every river in Asia discharges into the Pacific Ocean.*
2. *Nobody showed up to the party.*
3. *Somebody made a donation.* □

1.3 Truth Tables

Now we turn our attention to a fundamental means of keeping track of a statement's possible truth values.

A *truth table* is, loosely speaking, a table that lists all of the possible truth values of a statement, based strictly upon that statement's structure. For now, this description of what it means to be a truth table is informal. Over the next sections of the notes, we will sharpen our understanding of truth tables through practice. A formal definition of *truth table* is given later in Chapter 1.

Example 12. *The following truth table classifies how negations behave:*

P	$\neg P$
true	*false*
false	*true*

We read this truth table as follows. The top row (above the horizontal line) indicates that we are starting with some arbitrary statement named P, and we are going to investigate its negation $\neg P$: for each possible truth value of a statement P, what is the corresponding truth value of $\neg P$? Since P is a statement, P has only two possible truth values, true and false, and we check both values by listing them in the left column. The middle row (reading from the left to the right) indicates that if P is true, then $\neg P$ is correspondingly false. The bottom row (also reading from the left to the right) indicates that if P is false, then $\neg P$ is correspondingly true. □

Let me address an issue related to the choice and use of letters. Think about how, in an algebra class, we often use the letter x as the name of a number. Sometimes, we find x to be some specific number, even though x doesn't have to represent the same number from problem to problem. Other times, we don't know which number x might be, but we instead use the name x to discuss other related ideas. For example, even if x doesn't represent some specific number, you can still investigate the effects of squaring numbers by talking about x versus x^2.

Similar ideas occur in the study of logic. We often use the letter names P and Q to identify statements. Sometimes we might assign a specific statement to the name P, like we did in Example 5 when we let P be the statement "Babe Ruth hit exactly 714 home runs in his career." Other times, we may be talking about statements in more broad, abstract terms. For example, when we constructed the truth table in Example 12, we weren't thinking about Babe Ruth anymore; we were thinking about the broad effects of negating statements. In fact, when building any truth table, you are not thinking about some specific statement or statements written as English sentences; you are instead thinking about how a statement's structure affects its truth value.

We will continute to investigate truth tables throughout this chapter.

Problem 13. *Complete the following truth table by writing one truth value in each of the four empty spots:*

P	$\neg P$	$\neg(\neg P)$
true		
false		

What is special about the truth values of the statement $\neg(\neg P)$, relative to the truth values of the statement P? [To answer this question, focus on describing what is special about the columns (for P and for $\neg(\neg P)$) themselves, and not on how you constructed those columns.] What conclusion(s) do you draw from this? □

1.4 "And," "Or," and Equivalence

To investigate more complicated statements, we now address two similar ways to join statements together. We will use the two very small, but important, words "and" and "or."

Definition 14. *Given two statements P and Q, define the* conjunction *of P and Q to be the statement "P and Q." This conjunction is denoted symbolically as $P \wedge Q$. We refer to P and Q as the* component statements *of the conjunction $P \wedge Q$, because a conjunction must be made up of such "component" pieces.* □

Problem 15. *Using the statements P and Q from Examples 5 and 6 as our component statements, the conjunction $P \wedge Q$ is the statement "Babe Ruth hit exactly 714 home runs in his career, and Hank Aaron hit exactly 800 home runs in his career."*

1. *Is that conjunction a true statement? Why or why not? For this part of the problem, you should use only the information given to you in the examples cited. [That is, go back and re-read Examples 5 and 6.]*
2. *Is the conjunction "Babe Ruth hit exactly 714 home runs in his career, and Hank Aaron hit exactly 755 home runs in his career" a true statement? Why or why not? For this part of the problem, feel free to do an internet search to find Hank Aaron's actual home run total.* □

Problem 16. *Complete the following truth table to classify the truth values of the conjunction. Notice here that we are starting with the arbitrary statement P and the arbitrary statement Q. Each of statement P and Q has two possible truth values, so there are four possible combinations of truth values for the pair: (1) P and Q each true, (2) P true, but Q false, (3) P false, but Q true, and (4) P and Q each false. That is why the following table has four rows of truth values to consider. Complete each row by writing down the correct truth value. [Consider your approach to Problem 15, and think about how many true components it should take to make a true conjunction.]*

P	Q	$P \wedge Q$
true	*true*	
true	*false*	
false	*true*	
false	*false*	

□

Problem 17. *Take the statement "Our dog eats chicken and beef." Rewrite this sentence as another actual sentence that looks more formally like a conjunction.* □

Definition 18. *Given two statements P and Q, define the* disjunction *of P and Q to be the statement "P or Q." This disjunction is denoted symbolically as $P \vee Q$. As we did in Definition 14, we refer to P and Q as the* component statements *of the disjunction $P \vee Q$.* □

To help you remember the difference between the symbols $\wedge$ and $\vee$, notice that $\wedge$ looks like an A, which stands for "and." [Note, though, that this resemblance is simply a coincidence.]

Example 19. *Using the statements P and Q from Examples 5 and 6 as our component statements, the disjunction $P \vee Q$ is the statement "Babe Ruth hit exactly 714 home runs in his career, or Hank Aaron hit exactly 800 home runs in his career." This disjunction is a true statement, since despite the fact that Hank Aaron did not actually hit exactly 800 home runs in his career, it is still the case that Babe Ruth did actually hit exactly 714 home runs in his career.* □

Problem 20. *In the "Homer the Great" episode of* The Simpsons*, Homer learns that in order to join the Stonecutters (a secret club with a catchy theme song), you have to be the son of a Stonecutter or save the life of a Stonecutter. Suppose that you are the son of a Stonecutter and you save the life of a Stonecutter. In this situation, should you be allowed to join the Stonecutters? Briefly explain why or why not.* □

Problem 21. *Complete the following truth table to classify the truth values of the disjunction. [Consider the reasoning of Example 19 and your approach to Problem 20, and think about how many true components it should take to make a true disjunction.]*

P	Q	$P \vee Q$
true	*true*	
true	*false*	
false	*true*	
false	*false*	

□

Definition 22. *We refer to any statement that includes a negation, a conjunction, or a disjunction as a* compound statement. *We refer to negations, conjunctions, and disjunctions as* (logical) operations *in the sense that we use these operations to construct more complicated compound statements out of simpler component statements.* □

Warning: In a grammar course, we would consider a sentence to be a compound sentence if it consists of two complete sentences joined by the word "and" or by the word "or" or by some other acceptable connecting word. A sentence like "It is not raining" would not be considered a compound *sentence*, because it's not constructed from two or more complete sentences. However, the sentence "It is not raining" would actually be considered a compound *statement*, since it can be expressed as a negation of the simpler statement "It is raining." Therefore, in the context of a logic course, we do consider statements that include negations to be compound statements, even though that is not analogous to the approach in a grammar course. □

Before we proceed further, let me say a few words about constructing truth tables. In any truth table, we will start with some number of component statements, and we will operate on those component statements by using negations, conjunctions, or disjunctions to build compound statements out of the component statements. (We will learn more logical operations later in this chapter.) We will allow for all possible combinations of truth values for the component statements. We will then determine the truth value of the compound statements for each combination of truth values of the component statements. In our truth tables from this point, we will place all component statements on the left and all compound statements on the right, separating these two groups by a double vertical line. The truth values to the left of the double vertical line will completely dictate which truth values belong to the right of the double vertical line.

With the current vocabulary and standard practices in place, we now give a formal definition of *truth table*, as promised in the previous section.

Definition 23. *A* truth table *is a table that shows for each combination of truth values of some number of component statements, the corresponding truth value(s) of one or more compound statements which involve any of the component statements.* □

Problem 24. *Complete each truth table:*

P	$\neg P$	$P \vee (\neg P)$
true		
false		

P	$\neg P$	$P \wedge (\neg P)$
true		
false		

What is special about the truth values of the statement $P \vee (\neg P)$? What is special about the truth values of the statement $P \wedge (\neg P)$? [To answer these two questions, focus on describing what is special about the columns (for $P \vee (\neg P)$ and for $P \wedge (\neg P)$) themselves, and not on how you constructed those columns.] □

Warning: On every truth table that you complete, you are to show one column for each logical operation that is involved. This will count as "showing your work." For example, if you are supposed to complete a truth table for the statement $(P \wedge Q) \vee (\neg Q)$, then you should have columns for P, Q, $P \wedge Q$, $\neg Q$, and, finally, $(P \wedge Q) \vee (\neg Q)$. □

The next definition formalizes the important concept of being able to say the same thing in many different ways.

Definition 25. *Let R and S be statements, where (usually) at least one of these two statements is a compound statement. We say that R and S are* (logically) equivalent *to each other if, for each combination of truth values of the component statements that may make up R and S, we have that R is true when S is true and that R is false when S is false. We write $R \equiv S$ as shorthand for the sentence "R is equivalent to S."* □

Definition 25 allows us to identify two statements in a truth table as being logically equivalent to each other when the column of truth values underneath one statement exactly matches the column of truth values underneath the other statement. When two statements are equivalent, they are effectively interchangeable because they have the exact same meaning (by virtue of being true under the same set of circumstances).

Warning: In light of the previous paragraph, if a problem asks you to construct a truth table in order to determine whether/why two statements are equivalent, and if the problem asks you to briefly explain your choice (why there is or is not equivalence), then your brief explanation underneath the truth table simply amounts to saying (in one sentence) whether or not the two relevant columns match. □

Example 26. *Recall Problem 13, where we saw the truth table*

P	$\neg P$	$\neg(\neg P)$
true	*false*	*true*
false	*true*	*false*

Because the columns for P and $\neg(\neg P)$ match each other, we can say that $P \equiv \neg(\neg P)$. □

Example 27. *Let P be the specific statement "I am able to juggle." Then $\neg P$ can be cleverly phrased as "I am unable to juggle" and $\neg(\neg P)$ as "I am not unable to juggle." Based on the equivalence shown in Example 26, we see that our specific statements "I am able to juggle" and "I am not unable to juggle" are making the same assertion, even though the second is certainly phrased in a more awkward manner. If I actually can juggle, then P and $\neg(\neg P)$ are asserting the same correct fact. If I actually can't juggle, then P and $\neg(\neg P)$ are telling the same lie.* □

Problem 28. *Complete the truth table:*

P	Q	$P \wedge Q$	$(P \wedge Q) \vee Q$
true	*true*		
true	*false*		
false	*true*		
false	*false*		

Use your truth table to identify a simpler statement that is equivalent to the statement $(P \wedge Q) \vee Q$, and explain your choice in one brief sentence. □

Problem 29. *Use the equivalence that you discovered in Problem 28 to write a shorter sentence that is equivalent to "It is raining and it is snowing, or it is snowing."* □

Problem 30. *Complete the following truth table. Note that we use three component statements P, Q, and R, so there are eight combinations of truth values to consider. The eight rows of truth value entries represent those eight combinations.*

P	Q	R	$P \vee R$	$Q \wedge R$	$(P \vee R) \vee (Q \wedge R)$
true	*true*	*true*			
true	*true*	*false*			
true	*false*	*true*			
true	*false*	*false*			
false	*true*	*true*			
false	*true*	*false*			
false	*false*	*true*			
false	*false*	*false*			

Is the compound statement $(P \vee R) \vee (Q \wedge R)$ logically equivalent to any simpler statement in this table? If so, which one? If not, why not? In either event, explain your choice in one brief sentence. □

Problem 31. *Demonstrate the following equivalences by constructing a truth table. After each, write one brief sentence that indicates why the requested pairs of statements are equivalent.*

1. $P \wedge Q \equiv Q \wedge P$
2. $P \vee Q \equiv Q \vee P$ □

This previous problem tells us that when forming the conjunction or the disjunction of two component statements, the order of the component statements does not affect the truth value of the conjunction or the disjunction.

The next three problems deal with the important issue of negating conjunctions and disjunctions. If someone makes a claim that uses the word "and" or the word "or," then these problems show you how to take the appropriate opposing viewpoint every time.

Problem 32. *Construct a truth table to show that $\neg(P \wedge Q) \equiv (\neg P) \vee (\neg Q)$. After the table, write one brief sentence that indicates why the requested pair of statements is equivalent.* □

Problem 33. *I make the claim "It will rain or it will snow," but you don't agree with me. What would you say to me in order to declare your disagreement? In other words, what do you think the negation of "It will rain or it will snow" is? Phrase your answer as a compound sentence, using the words "rain" and "snow." Explain your reasoning. [Think about how many false components it takes to make a false disjunction.]* □

Note that in Prolem 33, I asked you to negate a specific disjunction. In Problem 34 below, I am asking you to generalize the result of Problem 33: instead of asking you how to negate a specific disjunction, I am asking how you would *ever* negate *any* disjunction.

Problem 34. *Considering the underlying structure of your response to the previous problem, supply a statement that is equivalent to* $\neg(P \vee Q)$. *[Your answer to this question should involve the letters P and Q and the correct logical operation symbols, not specific references to weather.] Support your answer by constructing a truth table that contains a column for* $\neg(P \vee Q)$, *a column (for the appropriate other statement) whose truth values match that column, and columns for any intermediate statements (considering the warning above Definition 25). After the table, write one brief sentence that indicates why the relevant pair of statements is equivalent.* □

Warning: Do not attempt the next two problems without completing Problems 32 and 34. □

Problem 35. *Write the negation of each of the following two statements, and for each, cite a previous problem that illustrates why you responded as you did. Each negation should be a compound sentence.*

1. *The radiator needs to be replaced, and the engine mounts need to be replaced.*
2. *Thanksgiving is on a Monday, or Thanksgiving is on a Tuesday.* □

Problem 36. *Write the negation of each of the following two statements, and for each, cite a previous problem that illustrates why you responded as you did. Each negation should be a compound sentence.*

1. *The meeting will be rescheduled, or the meeting will be canceled.*
2. *The defendant did threaten his blackmailer, but the defendant did not commit murder.* □

Theme: Any problem you have already completed is now a tool in your kit to help you complete the problem at hand. For example, completing Problems 35 and 36 is made possible by completing *and recalling* Problems 32 and 34. In general, I will allow you to cite the result of any problem in the notes prior to the problem you're completing, but not the result of any problem that comes after the problem you're completing. □

Problem 37. *Use a truth table to show that* $P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$. *After the table, write one brief sentence that indicates why the requested pair of statements is equivalent.* □

Problem 38. *Would the equivalence given in Problem 37 still hold if every conjunction symbol is replaced with a disjunction symbol, and vice versa?*

Show your work by constructing a truth table. After the table, write one brief sentence that indicates why the relevant pair of statements is or is not equivalent. □

Problem 39. *Supply an equivalence that hasn't been mentioned in the notes so far. I'm asking you to play around with our logical operations and identify two statements whose truth table columns match each other (so you must construct a truth table). Then, use your findings to write an example of two specific statements (that is, two complete sentences that are probably compound sentences) that are phrased differently, but mean the same thing.* □

Problem 40. *When we talk about two statements that are equivalent to each other, we use a three-bar symbol* ($\equiv$) *and not a familiar equals sign* ($=$). *What do you think it might mean for two statements to be* equal *to each other, as opposed to being* equivalent *to each other?* □

1.5 Conditionals

Have you ever felt like someone broke a promise they made to you? Maybe they didn't. In this section, you will see if your feelings were justified, because we will discuss a powerful and frequently misunderstood logical statement that is the life-blood of the mathematics community: the *conditional*.

Definition 41. *Given two statements P and Q, define the* conditional $P \to Q$ *to be the statement "If P, then Q." As we did in the previous section, we refer to the conditional $P \to Q$ as a compound statement, which is made up of the component statements P and Q. We call the statement P the* hypothesis *and the statement Q the* conclusion*, borrowing terminology reminiscent of science class. Amend Definition 22 to include the conditional as another type of logical operation.* □

Warning: Note that, without any mathematical context, the word "conditional" sounds like an adjective. However, if you re-read the previous definition, then you will see that "conditional" is used as a noun. □

Problem 42. *Take the conditional "If Rex barks at you, then he's hungry." What is the hypothesis of this conditional? What is the conclusion?* □

Example 43. *Let P be the statement "My fever reaches 105 degrees," and let Q be the statement "I go to the hospital." The conditional $P \to Q$ is the statement "If my fever reaches 105 degrees, then I go to the hospital."*

Let's investigate the possible truth values of the statement $P \to Q$ from this example. Imagine that I have made you that promise: "If my fever reaches 105 degrees, then I go to the hospital."

1. *If my fever does reach 105 degrees and I do go to the hospital, then I have kept my promise. The hypothesis would be true and so would the conclusion; I have told the truth.*

2. *If my fever does not reach 105 degrees and I do not go the hospital, then I have kept my promise (by not breaking it). The hypothesis would be false, and so would the conclusion; I have told the truth.*

3. *If my fever does not reach 105 degrees, but I go to the hospital anyway, then* I have not broken my promise. *I could have gone to the hospital for any other reason. Notice that I did not promise that if my fever did not reach 105 degrees, then I would not go to the hospital. (It is a common misunderstanding of conditionals to assume that this related promise has been made.) The hypothesis would be false, but the conclusion would be true; I have told the truth.*

4. *The only way that I can break my promise is if my fever does reach 105 degrees, but I do not go to the hospital. The hypothesis would be true, but the conclusion would be false; I have not told the truth.* □

Let's equate a promise kept with a true conditional, and let's equate a promise broken with a false conditional. We can then use the reasoning in the above example to complete the truth table for the conditional $P \to Q$, where P and Q are now completely arbitrary statements:

P	Q	$P \to Q$
true	true	true
true	false	false
false	true	true
false	false	true

Warning: Digesting this table, particularly the second and third rows of truth values, may very well be the most important task in this course. Using the same type of "promise" language as before, it is critical to remember that the only way to break an "if-then" promise is for the hypothesis to be true while the conclusion is false. A false hypothesis always results in a true conditional: the promise is kept by virtue of the promise-maker not being "on the hook" to follow up on the promise. □

Problem 44. *Complete the following truth table:*

P	$P \to P$
true	
false	

What is so special about the column of truth values under $P \to P$? [To answer this question, focus on describing what is special about the column itself, and not on how you constructed that column.] Write a specific simple statement (as a sentence), and call that statement P. Using that specific *statement P, write the statement $P \to P$, as a sentence. What is the truth value of the sentence $P \to P$ that you constructed?* □

Problem 45. *Complete the following truth table:*

P	Q	$\neg P$	$\neg Q$	$(\neg P) \to (\neg Q)$	$P \to Q$
true	*true*				
true	*false*				
false	*true*				
false	*false*				

Is $P \to Q$ equivalent to $(\neg P) \to (\neg Q)$? Explain your choice in one brief sentence. After explaining, go back and re-read all of Example 43, and pay particular attention to Item 3. □

Problem 46. *Your grandfather claims, "If Jen made an A in history, then Grandma gave her fifty dollars," but you don't agree with him. What two things would have to happen in order to demonstrate that he is wrong? With those two things in mind, what do you think the negation of "If Jen made an A in history, then Grandma gave her fifty dollars" is? [I sincerely recommend that you imagine that you're actually voicing your disagreement to your grandfather instead of trying to sound fancy for a question in a math class].* □

We now generalize the previous problem.

Problem 47. *Considering the underlying structure of your response to the previous problem, supply a statement that is equivalent to $\neg(P \to Q)$. [Your answer to this question should involve the letters P and Q and the correct logical operation symbols, not specific references to family members, history grades, or money.] Support your answer by constructing a truth table that contains a column for $\neg(P \to Q)$, a column (for the appropriate other statement) whose truth values match that column, and columns for any intermediate statements (considering the warning above Definition 25). After the table, write one brief sentence that indicates why the relevant pair of statements is equivalent.* □

Problem 48. *Write the negation of each of the following two statements. On both of these, you absolutely must refer to your answer from Problem 47, so you must complete that problem first.*

1. *If Sofia wins this round, then she wins the game.*
2. *If there is salt in the water, then the water boils faster.* □

Problem 49. *Suppose that I claim, "If Peggy Lee is dead, then Louis Armstrong is dead." A student confirms that Lee died in 2002 and that Armstrong died in 1971, and then claims, "Your claim is false. Lee's death couldn't have caused Armstrong's death. Armstrong died first!" However, my claim* *is actually correct. Using only what we've already seen about conditionals (thinking carefully about the hypothesis and the conclusion, in particular), briefly explain why my claim is indeed correct, despite the fact that Lee's death certainly didn't cause Armstrong's death. [Hint: the years of their deaths aren't what matters.]* □

Definition 50. *Let P and Q be statements. The conditional $Q \to P$ is called the* converse *of the conditional $P \to Q$.* □

Note that, by this definition, the converse of the conditional $Q \to P$ is the conditional $P \to Q$, because the converse of a conditional is constructed by simply swapping the hypothesis and conclusion of the given conditional. This means that the converse of a conditional's converse is simply that original conditional. For this reason, we can say that converses appear in pairs: $P \to Q$ and $Q \to P$ are converses of each other.

Problem 51. *Show that the conditional $P \to Q$ is not logically equivalent to its converse $Q \to P$. Use a truth table to show your work. After the table, write one brief sentence that indicates why the requested pair of statements is not equivalent.* □

Problem 52. *Write the converse of the conditional "If I have a runny nose, then I cancel class." Next, look at your truth table from Problem 51 (but don't recopy it here) to fill in the blanks with truth values: If "I have a runny nose" is (blank) and if "I cancel class" is (blank), then the given conditional would be true, but its converse would be false.* □

Problem 53. *Suppose that a homework problem asks you to write the converse of the conditional "If I cash in my chips, then I got a royal flush." Suppose that a particular student's response is "If I get a royal flush, then I will cash in my chips." This student's response is not correct. Briefly explain why, by identifying the error and offering a correction. [Hint: the truth values of any of the statements involved aren't what matters.]* □

Definition 54. *Let P and Q be statements. The conditional $(\neg Q) \to (\neg P)$ is called the* contrapositive *of the conditional $P \to Q$.* □

Problem 55. *Write an example of a conditional (as a complete sentence), and also write its contrapositive.* □

Problem 56. *Is the conditional $P \to Q$ logically equivalent to its contrapositive $(\neg Q) \to (\neg P)$? Use a truth table to show your work. After the table, write one brief sentence that indicates why the requested pair of statements is or is not equivalent.* □

Problem 57. *Take the conditional "If Chicago is freezing, then Milwaukee is freezing." Write a statement that is equivalent to that conditional. [The given sentence is certainly equivalent to itself, but devise a more substantial answer than that.] Justify your answer by citing a previous relevant problem.* □

1.6 Biconditionals

The following variation of Definition 41 creates a stronger type of promise than the ones we've already encountered.

Definition 58. *Given two statements P and Q, define the* biconditional $P \leftrightarrow Q$ *to be the statement "P if and only if Q." We take this to mean that $P \leftrightarrow Q$ is a shorthand way of notating the compound statement $(P \to Q) \wedge (Q \to P)$. Said another way, we take $P \leftrightarrow Q$ to be equivalent to $(P \to Q) \wedge (Q \to P)$. Given our previous vocabulary, $P \leftrightarrow Q$ is a compound statement, made up of component statements P and Q, and the biconditional is yet another type of logical operation.* □

Example 59. *The biconditional "I go to the gym if and only if it's Thursday" can be alternately written as "If I go to the gym, then it's Thursday, and if it's Thursday, then I go to the gym."* □

Let's take a look at the following truth table, noting that the last column is completed by taking the conjunction of the statements in the previous two columns:

P	Q	$P \to Q$	$Q \to P$	$P \leftrightarrow Q$
true	true	true	true	true
true	false	false	true	false
false	true	true	false	false
false	false	true	true	true

Notice from this table that for $P \leftrightarrow Q$ to be a true statement, we must have that the component statements P and Q have the same truth value as each other (that is, P and Q are both true or both false).

Problem 60. *Refer back to Example 43, where we saw the conditional "If my fever reaches 105 degrees, then I go to the hospital." Write the biconditional (using the phrase "if and only if") that has the same component statements as that conditional. Next, look at the truth table after Example 59 (but don't recopy it here) to fill in the blanks with truth values: If "My fever reaches 105 degrees" is (blank) and if "I go to the hospital" is (blank), then the given conditional would be true, but the corresponding biconditional would be false.* □

Problem 61. *Complete the following truth table:*

P	Q	$P \leftrightarrow Q$	$(P \leftrightarrow Q) \wedge Q$
true	*true*		
true	*false*		
false	*true*		
false	*false*		

We have seen a simpler statement before that is equivalent to $(P \leftrightarrow Q) \wedge Q$. What is that simpler statement? □

Problem 62. *Use a truth table to show that* $\neg(P \leftrightarrow Q) \equiv [P \wedge (\neg Q)] \vee [Q \wedge (\neg P)]$. *After the table, write one brief sentence that indicates why the requested pair of statements is equivalent.* □

Problem 63. *Use the previous problem to write the negation of the biconditional "It is raining if and only if it is thundering."* □

1.7 Variable Statements and the words "All" and "Some"

So far, we have investigated the truth values of some specific, isolated statements, as well as the truth values of unspecified statements that had specified structures. In this section, we begin to expand our collection of relevant types of statements.

Have you ever heard someone use the variable x as a placeholder in a spoken comment? "It will take x days to complete the first task." The truth value of such a statement depends on which x you consider. While this is sometimes frustrating in conversation ("Just tell me how many days!"), it is very useful in mathematics.

Definition 64. *We call a phrase* $P(x)$ *(read out loud as "P of x") a* variable statement *if, whenever a specific singular noun replaces x in the phrase, the resulting phrase is a statement. That is,* $P(x)$ *is a complete declarative sentence that is true or false (but not both) for each x that we choose. We refer to x as the* variable *in the variable statement* $P(x)$. □

Problem 65. *Let* $P(x)$ *be the variable statement "x is a U.S. state." Write each of the following three statements as sentences. Which of these statements are true? Which are false? Do some online fact-checking if you don't recall the U.S. states.*

1. $P(\textit{North Carolina})$
2. $P(\textit{Miami})$
3. $P(\textit{soccer})$ □

Problem 66. *Create (and write down) a sentence that is a variable statement, but has two variables: x and y. Next, insert specific singular nouns for each variable, write the resulting statement (as a sentence), and determine the truth value of that resulting statement. Be sure to pick a statement whose truth value can be determined easily by the class.* □

The following defines a type of statement that shows up in many mathematical discussions.

Definition 67. *A* universal conditional *is a statement that is written in the form "For all x, if* $P(x)$*, then* $Q(x)$*," where* $P(x)$ *and* $Q(x)$ *are variable* *statements which involve the same variable x. In symbols, such a universal conditional could be written as "For all x, $P(x) \to Q(x)$."* ☐

Example 68. *The following statements are universal conditionals. Both of these statements happen to be true.*

1. *For all x, if x is an integer, then $2 \cdot x$ is an even number.*
2. *For all x, if x is an insect, then x does not have a backbone.* ☐

Warning: While a universal conditional will always include the mention of a variable, universal conditionals are not technically variable statements themselves. Take, for example, the first universal conditional from Example 68 above: "For all x, if x is an integer, then $2 \cdot x$ is an even number." If we substitute a specific x (for example, let x be 8), then the sentence would become "For all 8, if 8 is an integer, then $2 \cdot 8$ is an even number," which is grammatical nonsense, and therefore, not a statement (see Definition 64). The issue here is that universal conditionals don't allow us to plug in specific singular nouns for x because they address all x-values simultaneously. ☐

Look back at the first item in Problem 11, where we saw the statement "Every river in Asia discharges into the Pacific Ocean." We did not explicitly use variables at that point, but we can now rephrase this statement as the universal conditional "For all x, if x is a river in Asia, then x discharges into the Pacific Ocean." As we saw in that problem, an acceptable way to write the negation is "There is at least one river in Asia that does not discharge into the Pacific Ocean." However, that negation can be rephrased to include the mention of a variable: "For some x, x is a river in Asia, and x does not discharge into the Pacific Ocean." We call such a statement an *existential statement*.

Definition 69. *An* existential statement *is a statement of the form "For some x, $P(x)$," where $P(x)$ is a variable statement (usually also a compound statement).* ☐

Problem 70. *Complete each of the following.*

1. *Write the statement "All cows eat grass" as a universal conditional.*
2. *Write the statement "Some person eats cheese" as an existential statement. [Your answer should include a conjunction, not a conditional.]*
   ☐

Prior to Definition 69, we saw how to negate a specific universal conditional. In general, given a universal conditional "For all x, if $P(x)$, then $Q(x)$," we can apply the same intuition that allowed us to answer Problem 11 in order to see that the negation would be an existential statement: "For some x, the statement 'if $P(x)$, then $Q(x)$' is false." This means that we have to consider the negation of the conditional $P(x) \to Q(x)$, but Problem 47 showed us that the negation of such a conditional is the conjunction $P(x) \wedge [\neg Q(x)]$. Therefore, the negation of the universal conditional "For all x, if $P(x)$, then $Q(x)$" is the existential statement "For some x, $P(x)$ and $\neg Q(x)$."

Problem 71. *Write the negation of "All cows eat grass" as an existential statement. Write the negation of "Some person eats cheese" as a universal conditional.* □

Problem 72. *Write the statement "No dogs eat caviar" as a universal conditional. Write its negation as an existential statement.* □

1.8 Problems

Here are some spare problems to close out the chapter.

Problem 73. *Write the negation of the sentence "I play the lottery, but I do not win," but write the negation as a conditional instead of a disjunction.*

Problem 74. *Write a complete declarative sentence (without variables) that is not a statement. Briefly explain your reasoning.* □

Problem 75. *The sentence "It will snow at LAX Airport on January 3, 2095" is a prediction. Is that prediction a statement? Briefly explain your reasoning.* □

Problem 76. *Supply a compound statement that is true for all possible truth values of its component statements (a compound statement that hasn't been mentioned in the notes so far). I'm asking you to play around with our logical operations and identify a statement whose truth table column has no false values (so you must construct a truth table). Then, use your findings to write an example of a specific statement (as a complete sentence) that is true under all circumstances.* □

Problem 77. *In Problem 62, you were asked to use a truth table to show that $\neg(P \leftrightarrow Q) \equiv [P \wedge (\neg Q)] \vee [Q \wedge (\neg P)]$. Try to establish that same result, but instead of using a truth table, use the fact from Definition 58 that $(P \leftrightarrow Q) \equiv [(P \to Q) \wedge (Q \to P)]$, and use the equivalences from Problems 32 and 47. Justify each step of your reasoning by citing the appropriate earlier problem or definition.* □

1.9 Chapter Summary

We update Definition 22 to say that a *compound statement* is any statement which includes a negation, conjunction, disjunction, conditional, or biconditional. We will also say that our list of *logical operations* now includes negations, conjunctions, disjunctions, conditionals, and biconditionals.

The following are truth tables for the logical operations that we have covered: negation, conjunction, disjunction, conditional, and biconditional.

P	$\neg P$
true	false
false	true

P	Q	$P \wedge Q$
true	true	true
true	false	false
false	true	false
false	false	false

P	Q	$P \vee Q$
true	true	true
true	false	true
false	true	true
false	false	false

P	Q	$P \to Q$
true	true	true
true	false	false
false	true	true
false	false	true

P	Q	$P \leftrightarrow Q$
true	true	true
true	false	false
false	true	false
false	false	true

We also list a few key equivalences that we have seen:

1. Any statement is equivalent to the negation of its negation: $P \equiv \neg(\neg P)$.
2. Conjunctions can be written in any order: $P \wedge Q \equiv Q \wedge P$.
3. Disjunctions can be written in any order: $P \vee Q \equiv Q \vee P$.
4. The negation of a conjunction is the disjunction of the negations of the component statements: $\neg(P \wedge Q) \equiv (\neg P) \vee (\neg Q)$.
5. The negation of a disjunction is the conjunction of the negations of the component statements: $\neg(P \vee Q) \equiv (\neg P) \wedge (\neg Q)$.
6. The negation of a conditional is the conjunction of the hypothesis and the negation of the conclusion: $\neg(P \to Q) \equiv P \wedge (\neg Q)$.
7. Any conditional is equivalent to its contrapositive:
   $P \to Q \equiv (\neg Q) \to (\neg P)$.

8. A biconditional is defined as the conjunction of a conditional and that conditional's converse: $P \leftrightarrow Q \equiv (P \to Q) \wedge (Q \to P)$.

9. The negation of a biconditional is the disjunction of two conjunctions, each of which contains one of the given biconditional's components and the negation of the other component:
$\neg(P \leftrightarrow Q) \equiv [P \wedge (\neg Q)] \vee [Q \wedge (\neg P)]$.

10. The negation of the universal conditional "For all x, if $P(x)$, then $Q(x)$" is the existential statement "For some x, $P(x)$ and $\neg Q(x)$."

Chapter 2

Proof

In this chapter, we take the opportunity to agree on some terminology surrounding the mathematical concept of *proof*. Some of these terms will be familiar to you, for better or worse. That is, some familiar terms may mean one thing in everyday conversation and mean something slightly different in a mathematical context. Pay attention to these distinctions. Armed with this terminology, we will investigate and practice various methods of proof.

2.1 Definitions and Axioms

In this section, we address two main ingredients of mathematical discussion: definitions and axioms. We begin with the head-scratching task of defining the word "definition." The following is a variation of a Merriam-Webster entry.

Definition 78. *A* definition *is a declaration of the meaning of a word or word group or a sign or symbol.* □

Example 79. *The whole point of defining a word, word group, sign, or symbol is to recognize a property or action that is distinct from all other properties or actions. Take, for example, Merriam-Webster's definition of "toffee": candy of brittle but tender texture made by boiling sugar and butter together. In stating this definition, an inherent equivalence is being declared. Namely, a thing is called "toffee" if and only if that thing is a candy of brittle but tender texture made by boiling sugar and butter together. Consequently, anything that is a candy of brittle but tender texture made by boiling sugar and butter together is called "toffee," and anything that is not a candy of brittle but tender texture made by boiling sugar and butter together is not called "toffee." The word "toffee" and its description are interchangeable. Definitions will always behave this way; going by a special name is the same as having certain agreed-upon properties.* □

If we were to open a standard dictionary, then we would find that we could read a definition of some word. That definition itself would contain words whose definitions we could find and read, and we could find and read the definitions of *those* words, and so on. We could eventually hop our way around the dictionary in this manner, perhaps finding our way back to the word with which we started. In mathematics, this does not and can not happen. In order to ensure an airtight understanding of what a new word means, that new word's definition would have to consist entirely of words whose meanings have already been illustrated ("older words," in a sense). However, since only finitely many words exist, we cannot continue forever saying that "this old word is defined by these older words, and those older words are defined by some even older words, and those even older words..." In order to actually start a mathematical conversation, we have to admit that some words that we use won't be defined in terms of "older words." This situation is described in the first part of the following definition.

Definition 80. *An* axiom system *is a collection of ideas consisting of the following pieces:*

1. *a list of terms that have no immediately specified meaning. These terms are called* undefined terms.
2. *a list of statements about the undefined terms. These statements are assumed to be true without any formal justification. These statements are called* axioms. □

Example 81. *Here is a common axiom system that relates to geometry. The undefined terms are* point*,* line*, and* belongs to*. Take the following three statements as axioms.*

1. *For each line, there are at least two points that belong to that line.*
2. *For each line, there is at least one point that does not belong to that line.*
3. *There exists at least one line.* □

The three axioms in Example 81 (and any axioms in any axiom system) are neither scientifically nor mathematically verified, and they are not meant to be verified; they are simply declared, so as to start a mathematical discussion. The purpose of putting an axiom system in place is to say, "If we believe these statements (the axioms) to be true, then what would happen as a consequence?"

It may seem uncomfortable at this stage to admit undefined terms into a mathematical conversation–to use words that we are agreeing have no immediately specified meaning. However, re-read the paragraph above Definition 80 and also consider the following. If I were to dive deeply into the study of, say, geometry (by possibly adding more axioms to the system in Example 81 and by determining which statements would have to consequently be true), then through that study, I would obtain a sense of potential meaning for the terms that were undefined. For example, I would have notions of what it could mean to be a *line* by having derived many results about lines. In a sense, the terms that are undefined at the start of the conversation eventually obtain some meaning by having the conversation–by geniunely and deeply studying the axiom system and its consequences. This is true of each axiom system for any topic (not simply for Example 81).

We will not deeply study the axiom system from Example 81 in this course. However, in the final section of Chapter 2 (after you have had some practice in writing proofs), you will use the axiom system in Example 81 to prove one consequence of those axioms: that at least three points exist.

We will speak about other axiom systems throughout the rest of these notes, beginning with the next section.

2.2 An Axiom System for Real Numbers

In this section, we pause to apply some of the new ideas from Chapters 1 and 2 to a familiar mathematical setting: real numbers.

Consider an axiom system where the undefined terms are *real number*, *is less than*, *equals*, and *is greater than*. Let the notation "$x < y$" stand for the statement "x is less than y," let "$x = y$" stand for the statement "x equals y," and let "$x > y$" stand for the statement "x is greater than y."

Warning: We are listing *equals* as an undefined term here. The word doesn't technically have to mean anything right now, and we can't start the conversation without some words that don't mean anything right now (see the paragraph above Definition 80). However, every axiom that we will assume for real numbers and every result that we will derive from those axioms will agree with the way that most readers would interpret the word *equals*, through their lived experience in previous mathematics courses (see the second paragraph after Example 81). □

For this section of the notes, consider only these two formally stated axioms below, which involve multiple variables (see Problem 66 for the first mention of multiple variables).

Axiom 82. *For all x and y, if x and y are real numbers, then exactly one of the following three statements is true:*

1. $x < y$
2. $x = y$
3. $x > y$ □

Axiom 82 simply says that for any pair of real numbers that you select, the first will be less than the second or the first will equal the second or the first will be greater than the second, and only one of those three scenarios will be true.

Axiom 83. *For all real numbers x and y, $x < y$ if and only if $y > x$.* □

Axiom 83 simply says that *one real number's being less than a second real number* is the same as *the second's being greater than the first*. For example, the claims that $3 < 5$ and $5 > 3$ are actually the same claim.

For the remainder of this section, please feel free to use the ordering of real numbers that you recall from previous courses. For example, you are permitted to claim (without explanation) that $5 < 10$ and $-43.2 < 0$ and $900 > 2$, etc.

Problem 84. *Let $P(x)$ be the variable statement "$x < 7$." Write each of the following three statements, using the $<$ symbol. Which of these statements are true? Which are false?*

1. $P(1)$
2. $P(7)$
3. $P(10)$ □

Axiom 82 gives an indication of how to negate variable statements involving real numbers. For example, suppose that two real numbers have been selected: a named (yet otherwise unknown) real number x and the known real number 10. If I were to claim that $x < 10$, and if you wanted to dispute my claim, then Axiom 82 indicates that the only other possibilites are that $x = 10$ or that $x > 10$. Therefore, the negation of the variable statement "$x < 10$" is the variable statement "$x = 10$ or $x > 10$," which is commonly shortened to "$x \geq 10$" (read out loud as "x is greater than or equal to 10"). A similar notation $x \leq y$ is read out loud as "x is less than or equal to y" and stands for the statement "$x < y$ or $x = y$."

As a quick note before the next problem, the notation $x \neq y$ is read out loud as "x does not equal y" and stands for the statement "$x < y$ or $x > y$."

Problem 85. *Write the negation of each of the following statements. Phrase your negations by using whichever symbol ($<$, $>$, $=$, $\leq$, $\geq$, $\neq$) is appropriate.*

1. $x < y$
2. $5 > z$
3. $w = 3$ □

Problem 86. *Write the negation of each of the following statements. Phrase your negations by using whichever symbol ($<$, $>$, $=$, $\leq$, $\geq$, $\neq$) is appropriate.*

1. $x \leq 7$
2. $5 \geq 3$
3. $z \neq 2$ □

Definition 87. *An* inequality *is a statement of any of the following five forms: $x < y$, $x > y$, $x \leq y$, $x \geq y$, or $x \neq y$.* □

We close with one last special type of inequality and its negation.

Definition 88. *A* simultaneous inequality *is a statement of any of the following four forms: $x < y < z$, $x \leq y < z$, $x < y \leq z$, or $x \leq y \leq z$. The simultaneous inequality "$x < y < z$" stands for the conjunction "$x < y$ and $y < z$." The simultaneous inequality "$x \leq y < z$" stands for the conjunction "$x \leq y$ and $y < z$." The other two types of simultaneous inequalities stand for conjunctions that I will leave to the reader.* □

We could apply Axiom 83 to Definition 88 in order to state and use simultaneous inequalities of the form $x > y > z$ (where the greatest number is mentioned first instead of last), but we have relatively few remaining occurrences of simultaneous inequalities in these course notes, so let's use only the types of simultaneous inequalities that are given in the definition above.

Because simultaneous inequalities are conjunctions, Problem 32 reminds us that their negations are disjunctions.

Problem 89. *Write each given simultaneous inequality as a conjunction of two inequalities, and then write the negation of that conjunction.*

1. $3 < x < 7$
2. $0 < y \leq 5$ □

Problem 90. *Write the negation of the disjunction "$z < 8$ or $z \geq 19$," and then write that negation as a simultaneous inequality.* □

We will visit real numbers again heavily in Chapter 3 and briefly in Chapter 4, and we will visit inequalities again in Chapter 4. In particular, Chapter 3 will introduce more axioms (more than the two that were stated in this section) so that we can have a more robust conversation about real numbers. In no way was this section a complete treatment of real numbers.

2.3 Deduction

In this section, we will see a useful application of conditionals. *Deduction* is a fundamental way of asserting that a specific statement is true, as a consequence of knowing that other related statements are true.

Definition 91. *A* deduction *is an argument of the following form: if the conditional $P \to Q$ is known to be true, and if the hypothesis P is also known to be true, then the conclusion Q must consequently be true.* □

We can see that a deduction is an intuitively agreeable form of argument by investigating the following truth table, which was first seen after Example 43:

P	Q	$P \to Q$
true	true	true
true	false	false
false	true	true
false	false	true

The only row on which $P \to Q$ and P are both true is the top row. On that row, we see that Q is also true. Therefore, if $P \to Q$ and P are both true, then Q has to be true as well.

Example 92. *I tell my class, "If Friday turns out to be a snow day, then I will administer the test on Monday." Because I would never lie about such a thing, my conditional is a true statement. A snowstorm hits, and the university cancels Friday's classes, making the hypothesis "Friday turns out to be a snow day" a true statement. You can therefore deduce (that is, use deduction to figure out) that I will administer the test on Monday.* □

Problem 93. *I make you the faithful promise that if Friday turns out to be a snow day, then I will administer the test on Monday. You wind up spending all of Friday, Saturday, and Sunday at home trying to recover from the flu. You're so ill that you can't even summon the energy to find out if it snowed. You show up to class on Monday, where I administer the test. Seeing this, you smile and tell yourself that Friday was a snow day.*

Does this argument that Friday was a snow day have the same form as a deduction? Explain why or why not. Is it true that Friday must have been a snow day? Which row(s) on the truth table after Definition 91 support(s) your answer? □

2.4 Settling Conjectures

Once an axiom system is in place, those axioms can be used to start deducing which other statements must consequently be true.

Definition 94. *A* conjecture *is a statement (in the context of a given axiom system) whose truth value has not yet been determined. It is common to call a statement a conjecture if nobody knows its truth value, but it is also common for a single person to call a statement a conjecture if that person does not yet know its truth value.* To settle a conjecture *means to determine and justify its truth value.* □

Conjectures are the unsolved (but not necessarily unsolvable) mysteries of the mathematics world. A mathematician's research consists (more or less) of dreaming up conjectures (or seeking other mathematicians' conjectures) and trying to settle those conjectures. A common inner dialogue for a mathematician sounds like this: "Here's a statement that I think might be true. Is it? How can I show it?"

Example 95. *A lawyer/mathematician named Pierre de Fermat caused quite a stir by making the following conjecture (in the context of an axiom system that I will not describe here): there are no positive whole numbers x, y, and z that make the equation $x^n + y^n = z^n$ true when n is a whole number that is greater than* 2*. This conjecture surprisingly went unsettled for over* 300 *years.* □

It is important to note that many conjectures are phrased or can be phrased as universal conditionals.

Example 96. *The following conjecture (also from an axiom system that I will not describe here) is phrased as a universal conditional: for all N, if N is a number that is greater than zero, then there are two prime numbers larger than N whose difference is* 2*. This is, in fact, a fairly famous conjecture called "The Twin Prime Conjecture." As of May 14, 2022, this conjecture has not been settled, although progress has been made on variations of this conjecture.* □

Toward the very important task of settling conjectures, let's assess the circumstances under which a universal conditional is true or false.

As you have seen in the discussion immediately above Problem 71, the negation of the universal conditional "For all x, if $P(x)$, then $Q(x)$" is the existential statement "For some x, $P(x)$ and $\neg Q(x)$." The only way for "For some x, $P(x)$ and $\neg Q(x)$" to be true is for there to be at least one x for which $P(x)$ and $\neg Q(x)$ are both true. Such an x would make $P(x)$ true while making $Q(x)$ false. Therefore, since the only way for "For some x, $P(x)$ and $\neg Q(x)$" to be true is for there to be an x for which $P(x)$ is true and $Q(x)$ is false, and since "For all x, if $P(x)$, then $Q(x)$" and "For some x, $P(x)$ and $\neg Q(x)$" are negations of each other, we see that the only way for the universal conditional "For all x, if $P(x)$, then $Q(x)$" to be false is for there to be *some* x for which $P(x)$ is true and $Q(x)$ is false.

Definition 97. *Take the conjecture "For all x, if $P(x)$, then $Q(x)$". A specific x for which $P(x)$ is true and $Q(x)$ is false is called a* counterexample *to the conjecture. Note that the existence of a counterexample to the conjecture indicates that the conjecture is false.*

Warning: Re-reading Definition 97, please note that a counterexample is a singular noun, not a statement. □

Problem 98. *Take the conjecture "All native Texans are named 'Hank.'" [Since this conjecture isn't a deeply mathematical claim, let's not worry about placing this conjecture in the context of some axiom system.]*

1. *Rewrite this conjecture as a universal conditional.*
2. *Supply a counterexample to this conjecture, doing some online fact-checking, if needed. Briefly explain why your choice is a counterexample. [In light of the warning above this problem, you should phrase your answer as "(Blank) is a counterexample because...," where the blank is filled in by a singular noun, not a statement.]* □

In the previous problem, we see that the silly conjecture "All native Texans are named 'Hank'" is false because a counterexample exists. However, many sincerely-posed conjectures do turn out to be true. In fact, once a conjecture is posed, exactly one of three possible events occurs:

1. The conjecture remains unsettled forever, for some reason.
2. Someone determines that the conjecture is actually false. You wouldn't even have to call the conjecture a "conjecture" anymore; at that point, it's simply a false statement. When the conjecture is a universal conditional, finding a counterexample is the means of establishing the conjecture as false.
3. Someone determines that the conjecture is actually true. At that point in time, the statement in question is no longer called a conjecture; it is a *theorem*.

Definition 99. *A* theorem *is a statement that is shown to be true, in the context of a given axiom system.* □

Example 100. *In the context of an axiom system that I will not describe here, Pythagoras and his associates proved a famous theorem about right triangles: the sum of the squares of the legs of any right triangle is equal to the square of the hypotenuse. [Millions of people know this theorem simply as "$a^2 + b^2 = c^2$," but on its own, that phrasing fails to mention right triangles at all, much less which label is given to which side of the triangle.]*
□

Definition 101. *A* proof *of a theorem is a collection of statements aimed at asserting (beyond all doubt) the truth of that theorem. Each statement in the proof calls upon an axiom, definition, previous theorem, deduction, or (when the theorem is stated as a universal conditional) the theorem's hypothesis. Note that this list is neither exhaustive nor prescriptive; a proof may omit some of these ingredients, and a proof may use some other ingredients.* □

As mentioned in the previous definition, when a theorem is stated as a universal conditional, the proof of the theorem assumes (for free) that the hypothesis is true. To see why, we first remind ourselves of the truth table for conditionals.

P	Q	$P \to Q$
true	true	true
true	false	false
false	true	true
false	false	true

If we allow the component statements to be variable statements, then the table above becomes the following.

$P(x)$	$Q(x)$	$P(x) \to Q(x)$
true	true	true
true	false	false
false	true	true
false	false	true

Suppose that $P(x)$ is false for some particular x. If this is the case, then $P(x) \to Q(x)$ is true for that same x (the bottom two rows of the table say so). This is substantial; if I want to prove the universal conditional "For all x, if $P(x)$, then $Q(x)$," then the choices of x that make $P(x)$ false don't need to be addressed. Therefore, in order to prove that the universal conditional is true, I am free to assume that the hypothesis $P(x)$ is true. I am, however, on the hook for showing that $P(x)$ being true forces $Q(x)$ to be true as well.

Example 102. *Take the phrases* Team A, Team B, Team C, Team D, Team E, *and* is on *as undefined terms. Take the following four numbered statements as axioms.*

1. *For all x, if x is on Team A, then x is on Team B.*
2. *For all x, if x is on Team D, then x is on Team C.*
3. *For all x, if x is on Team A, then x is not on Team C.*
4. *For all x, if x is on Team B, then x is on Team E.*

We'll prove the following theorem: for all x, if x is on Team A, then x is not on Team D.

For any x that isn't on Team A, there is nothing to prove, so the proof of this specific universal conditional will assume and build on the statement that x is on Team A.

Proof. Assume that x is on Team A. By Axiom 3, we deduce that x is not on Team C. By the contrapositive of Axiom 2 (For all x, if x is not on Team C, then x is not on Team D), which we know from Problem 56 is equivalent to Axiom 2, we deduce that x is not on Team D. [Therefore, for all x, if x is on Team A, then x is not on Team D.] □

Problem 103. *Using the axiom system from Example 102, prove that for all x, if x is on Team A, then x is on Team E.* □

Warning: Tying together Definitions 94, 97, and 101, the instructions "settle the conjecture" indicate that if the conjecture can be phrased as a universal conditional, then you are to demonstrate either that the conjecture is true by writing a proof or that the conjecture is false by supplying a counterexample. These instructions will appear on dozens of problems throughout the rest of these notes (including the next problem), so incorporate this vocabulary quickly. □

Problem 104. *Settle the conjecture "For all x, if x is a Thursday, then x is Thanksgiving." That is, prove the conjecture to be true, or supply a counterexample (and briefly explain why your choice is a counterexample). Feel free to use online sources if you need to check information about dates. [Since this conjecture isn't a deeply mathematical claim, let's not worry about placing this conjecture in the context of some axiom system.]* □

Problem 105. *When a conjecture is phrased as a universal conditional, which is likely to be easier: proving a true conjecture to be true, or supplying a counterexample to a false conjecture? Briefly explain your choice.* □

Note that in order to prove a theorem, we will need to rely upon some form of previous results. Almost any new theorem (and there are new theorems being proven every day) has a proof that relies on previously proven theorems. Those older theorems are true because of even older theorems, and so on. However, there comes a time in that process of back-tracking when a result cannot be proven by pre-existing truths. At this stage, what you find is a collection of axioms: a list of accepted assumptions about the concepts to be studied. An axiom system must be in place before we can proceed to proving theorems; we must know the assumptions and the "rules of the game" before we can expand our list of associated facts through various techniques of proof. I ask the readers' pardon for having posed some examples and problems (like Example 96 and Problem 104) that did not refer to a specified axiom system, but in fairness, those examples and problems were designed to stand on their own, for the purpose of understanding some fragment of mathematical discourse. For instance, Problem 104 wasn't part of some larger effort to study days of the week or holidays; it was an exercise in settling a conjecture.

Before we address other proof techniques and problems, here is an aside on a word that is commonly misused. The following definition is from Merriam-Webster.

Definition 106. *The following are various accepted uses of the word* theory*:*

1. *the analysis of a set of facts in their relation to one another.*
2. *abstract thought: speculation.*
3. *the general or abstract principles of a body of fact, a science, or an art.*
4. *a belief, policy, or procedure proposed or followed as the basis of action.*
5. *an ideal or hypothetical set of facts, principles, or circumstances–often used in the phrase "in theory."*
6. *a plausible or scientifically acceptable general principle or body of principles offered to explain phenomena.*
7. *a hypothesis assumed for the sake of argument or investigation.*
8. *a body of theorems presenting a concise systematic view of a subject.* □

The word "theory" is much more common in conversations outside of mathematics, but the word "theorem" is much more common in conversations inside of mathematics. We will use the word "theorem" in this course far more than we will use the word "theory," so I want to note the distinction between these terms. They are not interchangeable terms.

Example 107. *The following are correct uses of the words "theorem" and "theory":*

1. *Knot theory is a field of study dealing with facts about knots.*
2. *I proved a couple of theorems about knots, and I put those results into an article.*
3. *I submitted that article to a journal that publishes papers in knot theory.* □

Please remember that a theorem is a statement whose truth has been proven, but a theory is essentially either a hunch on which you act (if you're speaking conversationally) or a field of study that is comprised of many theorems (if you're speaking mathematically). Therefore, mathematicians use the word "theory" in the sense of part 8 of Definition 106.

2.5 Proof by Contradiction

Is this section, we quickly address one significant proof technique.

Definition 108. Proof by contradiction *is a particular method of proving a statement to be true by showing that the statement can't be false: we assume that a statement that we want to prove to be true is instead false, and if some impossible result follows as a logical consequence, then the given statement can't possibly be false.* □

Warning: As an immediate consequence of Definition 108, if we are setting out to prove a statement by contradiction, then we must assume that statement's *negation*. Therefore, you must be mindful of the processes that we have seen for negating various types of statements. □

Example 109. *Let's prove the statement "There is no largest number" by contradiction. [Since this is a stand-alone example to illustrate a proof by contradiction, let's agree that this statement deals with real numbers, but let's not formalize what our axioms are. Therefore, if there is a useful fact from previous experience with algebra, then let's agree to use it on this example.]*

We first assume that the given statement is actually false, so we temporarily assume its negation: that there is *a largest number. Call that number N.*

The sum $N+1$ is a number, but $N+1$ is greater than N. [This is where we have applied previous experience with algebra. The next chapter of these notes features a robust and formal axiomatic development of such algebraic properties.] We therefore have that N is not the largest number.

We have now determined that N is the largest number and that N is also not the largest number, but this is a statement of the form $P \wedge (\neg P)$. We showed in Problem 24 that such a statement is false.

What we have proven is that if we assume the statement "There is no largest number" to be false, then we arrive at a situation that cannot occur, so "There is no largest number" must be true, so it is a theorem. □

Example 110. *If you're not familiar with Sudoku, then read a brief online explanation of the game's rules (which should be treated as axioms). You can use proof by contradiction as one way to master the game. For example, pretend that you have determined that a square must be filled in by either* *a* 1 *or a* 2*. If you temporarily assume that the square should be filled in by a* 1*, but if that choice causes you to break a rule of the game later, then you know that* 2 *was the right entry. [Sudoku gets called a "math game" for unfair reasons. Yes, Sudoku has numbers, but it's the* logic *involved that makes it a "math game."]* □

Problem 111. *Suppose that you wanted to prove the Calculus theorem "For all f, if f is a differentiable function, then f is a continuous function." If you wanted to prove this theorem by contradiction, then what very specific assumption [actually using the words "differentiable" and "continuous"] would you make in order to start your proof?* □

It's not inherently bad to prove a theorem by contradiction, but it is usually considered more elegant and more polite to avoid proof by contradiction, if it's possible.

Definition 112. *For our purposes, a proof that is not a proof by contradiction is called a* direct proof. □

Note that the proof in Example 102 is a direct proof.

2.6 A Final Word on Axioms

Before we proceed to the next chapters on axioms and theorems in number theory and in set theory, let's make an agreement regarding the philosophical structure of Chapter 1, in the light of what we've covered here in Chapter 2. In Chapter 2, we have essentially shown that there is a framework for trying to settle conjectures and to make rigorous arguments for a statement's truth. We must start with an axiom system, which is a list of our undefined terms and a list of our assumed rules (axioms) about those undefined terms. After the axiom system is in place, we use deduction and other techniques to prove a theorem, at which point, that first theorem becomes part of our arsenal for proving even more theorems.

Proving theorems is what develops the theory of any mathematical subject (see part 8 of Definition 106). The usefulness of the theory that is developed depends on many factors, but especially on how many theorems can be derived from the axioms and whether those theorems have applications in other branches of mathematics or in other disciplines. For instance, the axiom system in Example 102 was good for learning how to write a direct proof, but it does not yield a robust theory at all: we won't be able to prove many theorems from those four axioms, and those theorems won't give us a very deep understanding of what it means to be any of those teams. By contrast, Euclid started with five geometric axioms, and filled several *books* with theorems that followed from those axioms, yielding centuries of advancement in mathematics and in other disciplines. [Biographical sketches of Euclid and of David Hilbert are worth a read, if you can make the time, but I won't say more here.]

I should point out that the astute reader may very well ask, "Weren't we claiming to prove results back in Chapter 1? Did we declare any undefined terms or state any axioms about logic?" To be perfectly honest, we didn't start with an explicitly-stated axiom system. (Referencing the paragraph after Definition 1, we certainly could say that "complete declarative sentence" and "true" and "false" were undefined terms.) I purposely side-stepped an axiom system at the beginning in order to ease our entry into the subjects of logic and proof. Please be assured that we practiced tremendously useful mathematical thinking skills in Chapter 1, and our time and energy were not wasted. As we proceed into later chapters, we will treat our results from Chapters 1 and 2 as "axioms" and as "results whose truths have been established." With this in mind, we agree to call upon any result from Chapters 1 or 2 if it may help us to prove theorems and to settle conjectures in the upcoming chapters.

2.7 Exercises in Proof

In this section of the chapter, I pose some examples of axiom systems and ask many questions about their consequences. In many instances, I ask you to write proofs of theorems or to settle conjectures. With this second task in mind, please re-read Definition 94 and the Warning before Problem 104 to remind yourself of what it means to "settle a conjecture."

For the next three problems, use the following story and guidelines.

Janice and Floyd are packing for a trip together, and they are trying to plan some of their vacation activities based upon the items that they can pack in their carry-on luggage. [Assume that they have already packed their checked luggage (with other items that won't be mentioned below) and sent it ahead to the airport, so that no items can be added to or removed from their checked luggage.] Here are their guidelines for packing their carry-on luggage.

1. Each person has space to pack two items.
2. Exactly one person packs contact solution [Janice and Floyd both insist on wearing contacts, but they will share contact solution].
3. Janice packs a swimsuit if and only if Floyd packs a swimsuit [neither would consider swimming alone].
4. If one person packs a camera, then Janice or Floyd packs the camera's charger.
5. If Janice hikes, then Floyd packs a book to read [Floyd hates hiking].

6. Janice hikes on the trip if and only if she packs hiking shoes [the pair of shoes counts as a single extra item].

7. If Janice hikes on the trip, then she brings the camera on her hike [whether she packs it or Floyd packs it].

These seven guidelines for packing amount to axioms. That is, starting with these seven statements, we want to figure out more about Janice and Floyd's consequential actions and other restrictions. To have an axiom system, though, we also need undefined terms. Technically, *packing*, *trip*, *items*, *contact solution*, *swimsuit*, *camera*, and many other words and phrases in the list are undefined terms. For example, you don't need to attach any meaning to the word "camera" in order to prove any relevant theorems; you simply need to recognize the word itself and use it in a manner that agrees with the axioms provided. The same is true for all of the other undefined terms in this axiom system or in any other axiom system.

Problem 113. *Prove that if Floyd packs a swimsuit, then neither of them packs the camera. I suggest that you prove this by contradiction: show that a rule is broken if you assume the negation of "If Floyd packs a swimsuit, then neither of them packs the camera" (so you have to first determine that negation).* □

Problem 114. *Prove that Janice doesn't go hiking. I suggest that you prove this by contradiction: show that a rule is broken if you assume that Janice does go hiking.* □

Problem 115. *Is there any situation in which Floyd can pack a book? If yes, then list each person's packed items and show (one by one, in complete sentences) why each rule #1-7 is either not applicable or not broken. If not, then show which rule would be broken if Floyd packs a book, and show why.* □

For the next ten problems, use the following guidelines.

A hospital patient is put on a strict diet. There are only seven foods that he is ever allowed to eat: chicken, egg, bread, ham, broccoli, potato, and apple. The doctor prescribes the following rules for the patient's diet:

1. He must eat exactly four of the seven foods each day.

2. He must eat potato or bread (but not both) each day.

3. He must eat chicken or ham each day.

4. If he eats chicken and ham on the same day, then it does not happen that he eats both chicken and ham again the next day.

5. If he eats egg one day, then he does not eat chicken that day.

6. On each day, he eats apple if and only if he eats bread.
7. If he does not eat broccoli one day, then he does not eat egg the next day.
8. If he does not eat apple one day, then he eats apple the next day.
9. If he eats broccoli one day, then he does not eat broccoli the next day.

These nine rules for the patient's diet amount to axioms. Undefined terms include, but are not limited to *hospital patient*, *diet*, *foods*, *eat*, *chicken*, *egg*, etc.

Note that there is a variable hidden in these axioms: time. The choice of (somewhat) conversational language in the axioms avoids the use of x, but all of the axioms could be rewritten to explicitly include variables. For example, Axiom 1 could be rewritten as "For all x, he must eat exactly four of the seven foods on day x," Axiom 5 could be rewritten as "For all x, if he eats egg on day x, then he does not eat chicken on day x," and Axiom 9 could be rewritten as "For all x, if he eats broccoli on day x, then he does not eat broccoli on day $x+1$." However, please feel free to use the axioms exactly as they are phrased in the original list.

Recall from Problem 56 that a conditional is equivalent to its contrapositive, so we are free to use the contrapositives of any of these axioms (the ones that are conditionals) as if they were axioms themselves. Consider Axiom 5. If we swap the hypothesis for the conclusion and negate both of those component statements, we arrive at the conditional "If he eats chicken that day, then he does not eat egg one day." However, this sentence is not very clear; it references "that day" without first identifying what that day is. Therefore, a more useful version of the contrapositive of Axiom 5 is "If he eats chicken one day, then he does not eat egg that day." Call the first day that's mentioned "one day" and if you must later refer back to that same day, then call it "that day." Similar wordsmithing tackles the issue of rephrasing an axiom that references one day to the next. For example, a useful version of the contrapositive of Axiom 7 is "If he eats egg one day, then he eats broccoli the day before."

Problem 116. *In light of the above paragraph, state the contrapositives of Axioms* 8 *and* 9. □

On the remainder of these problems, if you use the contrapositive of an axiom, then cite it as the contrapositive of that axiom.

Problem 117. *Suppose that the patient's doctor reports to an assistant that the patient ate apple on some particular day. The assistant responds by saying, "By Axiom* 8*, the patient does not eat apple the next day." The assistant is not actually justified in making this claim because the assistant* *incorrectly believes that two certain statements are equivalent. Write down those two statements (as actual sentences that talk about apple). Find (and cite) a problem in Section* 1.5 *that shows that those two statements are not actually equivalent.* □

Problem 118. *Prove that if the patient eats potato one day, then he eats apple the next day. I suggest making this proof a direct proof.* □

Problem 119. *Prove that if the patient eats potato one day, then he eats ham that day. Feel free to try to prove this by contradiction.* □

Problem 120. *Prove that if the patient eats potato one day, then he does not eat broccoli the day before. Feel free to try to prove this by contradiction.* □

Problem 121. *Prove that if the patient eats potato one day, then he eats egg the next day.* □

Problem 122. *Prove that if the patient eats egg one day, then he does not eat broccoli that day.* □

Problem 123. *Prove by contradiction that if the patient eats egg one day, then he does not eat potato that day.* □

Problem 124. *Prove that if the patient does not eat ham one day, then he eats broccoli that day.* □

Problem 125. *Prove that if the patient does not eat ham one day, then he eats ham the next day.* □

For the next five problems, use the following guidelines.

The manager of a baseball team is trying to complete the lineup card, a task which consists of placing nine players in an ordered list, first through ninth. The nine players being considered for a particular lineup consist of three left-handed batters (the first baseman, the right fielder, and the pitcher) and six right-handed batters (the second baseman, the shortstop, the third baseman, the center fielder, the left fielder, and the catcher). Once all nine players have batted in order, the lineup starts over from the first batter, and continues as many times as necessary. For this reason, the first batter is considered the next batter after the ninth batter, even though the first batter is considered to bat earlier in the order than all of the other batters. Based on the strengths of the team, the manager has devised the following rules for creating a lineup:

1. No left-handed batter is allowed to be the next batter after another left-handed batter.
2. The shortstop or the right fielder must bat first.
3. The first baseman, the third baseman, or the right fielder must bat fourth.

4. If the first baseman does not bat fourth, then he bats third or fifth.
5. The pitcher must bat ninth.
6. If the opposing team's pitcher is right-handed, then a left-handed batter must bat fourth.
7. If the opposing team's pitcher is left-handed, then a right-handed batter must bat fourth.
8. The catcher must bat next in the order after the shortstop.
9. The center fielder bats either second or eighth.
10. The second baseman must bat later in the order than the right fielder.

Problem 126. *Prove by contradiction that the shortstop bats first.* □

Problem 127. *Prove that if the first baseman bats third, then the second baseman does not bat fifth.* □

Problem 128. *Prove that if the opposing team's picther is left-handed, then the left fielder does not bat third.* □

Problem 129. *Prove that if the opposing team's pitcher is right-handed, then the right fielder bats sixth.* □

Problem 130. *Is there a situation in which the second baseman can bat sixth? If yes, then list the lineup and show (one by one, in complete sentences) why each rule #1-10 is either not applicable or not broken. If not, then show which rule would be broken if the second baseman bats sixth, and show why.* □

2.8 Problems

Here are some spare problems to close out the chapter.

Problem 131. *Using only the axiom system described in Example 81 (the three undefined terms and the three axioms), prove that at least three points exist. [The statement to be proven is not a conditional, so your proof should not start with an assumption.]* □

Problem 132. *Based on your lived experience in a geometry course, how would you instinctively describe what it means to be a line? After you have written your description, consider whether your instincts caused you to consider drawings in a flat plane. Next, consider drawing instead on a globe. How would you describe what it means to be a line if you were drawing on that globe?* □

Chapter 3

Number Theory

In this chapter, we will use our formal techniques to prove some potentially familiar results about numbers. Based on Definition 106 and Example 107, "number theory" is the axiomatized study of numbers (most often, a special class of numbers called *integers*). Many questions in number theory deal with divisibility. Which numbers divide evenly into which other numbers? Which numbers are prime, that is, divisible only by trivial numbers? The Twin Prime Conjecture (Example 96) and the first item in Example 68 both deal with issues of number theory. One prominent modern application of number theory is cryptography; the security of a substantial amount of information depends on code-makers' ability to generate and manipulate incredibly large prime numbers. However, for the purposes of a gentler entry into the subject of number theory, we will concern ourselves with more basic and foundational issues.

3.1 Axioms and Algebraic Preliminaries

Recall from Definition 80 that, in order to start settling conjectures and writing proofs about numbers, we need some undefined terms and axioms in place, regardless of our previous prejudices about (or comfort level with) numbers. Please note that the following undefined terms and axioms are of my choosing and that there are other choices of undefined terms and axioms that would also generate a rich discussion about numbers. Regardless, please note that this rather familiar numerical content is simply one of many subjects in mathematics, and as such, is beholden to the same type of axiomatic structure that we have been discussing throughout these notes. Even in dealing with familiar numbers, we cannot escape the need to identify undefined terms, to agree upon some assumptions, and (when we write proofs) to supply reasons for our claims.

Take the word *integer* and the symbols $+$ and $\cdot$ as undefined terms. As we did in Section 2.2, take also *real number* and *equals* as undefined terms.

We can remind ourselves of Axioms 82 and 83, although we won't need them in this chapter.

Axiom 133. *Every integer is also a real number.* □

Warning: In Section 2.2, I foreshadowed the fact that even though we are technically using the word "equals" as an undefined term, every axiom that we will assume for real numbers and every result that we will derive from those axioms will agree with the way that most readers would interpret the word "equals," through their lived experience in previous mathematics courses. Similarly, most readers will have some idea of what an integer is; you could read ahead to Axiom 189 in order to confirm that you do indeed have some idea. Again, though, to start the conversation about integers, the word "integer" doesn't have to mean anything specific right now. It will be through the work of this chapter that we incrementally derive meaning for the word "integer," finally landing upon Axiom 189.

To elaborate on this last idea, we will indeed arrive at your familiar notions of any and all integers, but after encountering the specific integers 1, 0, 2, and -1, which happens after carefully developing those four integers' existence and behavior through axioms. Throughout this chapter, you will need to be careful to justify your reasoning by citing the given axioms, and not by relying on prior knowledge of and comfort with numbers. □

As mentioned in the introduction to this chapter, number theory deals more with integers than with other types of numbers. However, given Axiom 133, we will see that in order to uncover many truths about the integers, we first need to gain an understanding of the structure of the real numbers at large, and how the real numbers interact with the $+$ and $\cdot$ symbols. Such mathematical work is *algebraic* in nature: dealing with the structure of a set and with the behavior of some operations that are performed on that set. [We will talk about *sets* in depth in the next chapter. For now, treat *collection* as a suitable synonym for *set*.]

With this in mind, most of the rest of this section of the notes will deal with algebraic properties. A few results will deal with integers specifically, but the discussion about integers will take greater precedence in the next section of the notes.

Definition 134. *An* equation *is a statement of the form "$x = y$," where x and y are real numbers.* □

The next few axioms deal with the behavior of the $=$ symbol.

Axiom 135. *For all real numbers x and y, if $x = y$, then $y = x$ [if two real numbers are equal to each other, then the order in which you declare their equality does not matter].* □

Axiom 136. *For all x, if x is a real number, then $x = x$ [each real number equals itself].* □

Axiom 137. *For all real numbers x, y, and z, if $x = y$ and $y = z$, then $x = z$.* □

Later in this section, I will provide some guidance on how to use Axiom 137 concisely. For now, note that this axiom says that if some real number equals a second real number, which equals a third real number, then the first and third real numbers are equal to each other. In fact, this axiom can be used repeatedly to show that if some real number equals a second real number, which equals a third real number, which equals a fourth real number, which equals a fifth real number, ..., which equals a final real number, then the first and final real numbers are equal to each other.

These next few items deal with the behavior of the $+$ symbol.

Axiom 138. *For all x and y, if x and y are real numbers, then $x+y$ (read out loud as "x plus y") is a real number.* □

Here is a sharper version of the same axiom.

Axiom 139. *For all x and y, if x and y are integers, then $x+y$ is an integer.* □

Definition 140. *For real numbers x and y, the real number $x+y$ (whose existence is guaranteed by Axiom 138) is called the* sum *of x and y.* □

Problem 141. *Prove that for all x, y, and z, if x, y, and z are integers, then $(x+y)+z$ is an integer and $x+(y+z)$ is an integer.* □

Theorem 142. *For all x, y, and z, if x, y, and z are real numbers, then $(x+y)+z$ is a real number and $x+(y+z)$ is a real number.*

Proof. The proof of this theorem is the same as the proof that you will have provided in Problem 141, except that every appearance of "integer" should be replaced by "real number," and Axiom 138 should be used instead of Axiom 139. □

Axiom 143. *For all x, y, and z, if x, y, and z are real numbers, then the differently-grouped real numbers $(x+y)+z$ and $x+(y+z)$ from the previous theorem are equal to each other: $(x+y)+z = x+(y+z)$.* □

Going forward, feel free to interpret the ungrouped sum $x+y+z$ as $(x+y)+z$ or as $x+(y+z)$, which the previous axiom declares are equal to each other, anyway. Real numbers written as the sum of four or more other real numbers follow a similar, intuitive interpretation.

Axiom 144. *For all x and y, if x and y are real numbers, then $x+y = y+x$ [the order in which you write a sum does not matter].* □

Axiom 145. *There exists an integer 0 (read out loud as "zero") such that for all x, if x is a real number, then $x+0 = x$ (Axioms 144, 137, and 135 further guarantee that $0+x = x$, $x = x+0$, and $x = 0+x$).* □

Axiom 146. *For all x, if x is a real number, then there exists a real number $-x$ (read out loud as "negative x") such that $x+(-x)=0$.* □

Definition 147. *For a real number x, the real number $-x$ (whose existence is guaranteed by Axiom 146) is called an* additive inverse *of x.* □

The next few items deal with the $\cdot$ symbol.

Axiom 148. *For all x and y, if x and y are real numbers, then $x \cdot y$ (read out loud as "x times y") is a real number.* □

Here is a sharper version of the same axiom.

Axiom 149. *For all x and y, if x and y are integers, then $x \cdot y$ is an integer.* □

Definition 150. *For real numbers x and y, the real number $x \cdot y$ (whose existence is guaranteed by Axiom 148) is called the* product *of x and y.* □

Problem 151. *Prove that for all x, y, and z, if x, y, and z are integers, then $(x \cdot y) \cdot z$ is an integer and $x \cdot (y \cdot z)$ is an integer.* □

Theorem 152. *For all x, y, and z, if x, y, and z are real numbers, then $(x \cdot y) \cdot z$ is a real number and $x \cdot (y \cdot z)$ is a real number.*

Proof. The proof of this theorem is the same as the proof that you will have provided in Problem 151, except that every appearance of "integer" should be replaced by "real number," and Axiom 148 should be used instead of Axiom 149. □

Axiom 153. *For all x, y, and z, if x, y, and z are real numbers, then the differently-grouped real numbers $(x \cdot y) \cdot z$ and $x \cdot (y \cdot z)$ from the previous theorem are equal to each other: $(x \cdot y) \cdot z = x \cdot (y \cdot z)$.* □

Going forward, feel free to interpret the ungrouped product $x \cdot y \cdot z$ as $(x \cdot y) \cdot z$ or as $x \cdot (y \cdot z)$, which the previous axiom declares are equal to each other, anyway. Real numbers written as the product of four or more other real numbers follow a similar, intuitive interpretation.

Axiom 154. *For all x and y, if x and y are real numbers, then $x \cdot y = y \cdot x$ [the order in which you write a product does not matter].* □

Axiom 155. *There is an integer 1 (read out loud as "one") such that for all x, if x is a real number, then $x \cdot 1 = x$ (Axioms 154, 137, and 135 further guarantee that $1 \cdot x = x$, $x = x \cdot 1$, and $x = 1 \cdot x$).* □

The next two axioms reflect assumptions about numbers that you have surely used frequently (and to great utility) in algebra classes: adding or multiplying the same number on both sides of a correct equation maintains the correctness of that equation. In the proper language of our current axiom system, we write these assumptions as the following two axioms.

Axiom 156. *For all real numbers x, y, and z, if $x = y$, then $x + z = y + z$ (Axioms 144 and 137 further guarantee that if $x = y$, then $z + x = z + y$).* □

Axiom 157. *For all real numbers x, y, and z, if $x = y$, then $x \cdot z = y \cdot z$ (Axioms 154 and 137 further guarantee that if $x = y$, then $z \cdot x = z \cdot y$).* □

Recall from Problem 51 that a conditional and its converse are not equivalent. However, the following (which is the converse of Axiom 156) happens to be true.

Theorem 158. *For all real numbers x, y, and z, if $x + z = y + z$, then $x = y$.*

Proof. Assume that x, y, and z are real numbers for which

$$\begin{array}{rrcll} & x+z & = & y+z. & \\ \text{Then} & (x+z)+(-z) & = & (y+z)+(-z) & \text{by Axiom 156, so} \\ & x+(z+(-z)) & = & y+(z+(-z)) & \text{by Axioms 143 and 137, so} \\ & x+0 & = & y+0 & \text{by Axioms 146 and 137, so} \\ & x & = & y & \text{by Axioms 145 and 137.} \end{array}$$

□

Before we proceed to the next items, there are some style notes to address, regarding the proof of Theorem 158. The provided proof contains five lines that are equations. (The equals sign in each equation is aligned with the others, as a matter of neatness.)

The first of these five equations is the hypothesis that we assumed from the given claim. Notice the use of a few words prior to that first equation, though. We don't simply have variables immediately appearing in equations; we say what those variables are (they are real numbers) before we use them.

The second of these five equations is (as indicated) a consequence of Axiom 156, treating $x+z$ from the theorem as x from the axiom, $y+z$ from the theorem as y from the axiom, and $-z$ from the theorem as z from the axiom. All of this is perfectly legal; the variable names in any axiom are to be used as place-holders. (What you call x in any problem at hand may take the role of some quantity that goes by another name, as stated in a relevant axiom.) This is certainly worth stating as an aside right now, but does not require explicit mention in this (or any other) proof.

The third, fourth, and fifth of these five equations are (as indicated) consequences of Axiom 137, in addition to the other listed axioms. For example, the third equation is true because Axiom 143 guarantees that the left-hand side of the second equation equals the left-hand side of the third equation and that the right-hand side of the second equation equals the right-hand side of the third equation, at which point, Axiom 137 guarantees that the left- and right-hand sides of the third equation do indeed equal each other (by being equal to the same common value: either side of the second equation). (The use of Axiom 137 in moving to the fourth and fifth equations is similar.) However, this current paragraph would be a lot of extra information to include in the proof.

Therefore, in situations like these, where two axioms are being applied at the same time, but one axiom's usage is much more critical (or more interesting, frankly) than the other, most mathematicians will list the more critical axiom and omit the reference to the other axiom. In the above proof, that would amount to omitting every reference to Axiom 137. Ultimately, though, when you're in doubt about how much to detail to include or how many references to cite, over-explaining is more safe than under-explaining.

Overall, the format of the proof of Theorem 158 is useful in situations where we start by assuming that one equation is true, and are trying to prove that some other equation is true as a consequence of our assumption.

This next axiom deals with the interaction between the $+$ and $\cdot$ symbols.

Axiom 159. *For all x, y, and z, if x, y, and z are real numbers, then*

$$x \cdot (y+z) = (x \cdot y) + (x \cdot z)\,.$$

[Don't get hung up on the names x, y, and z. This axiom is saying that the product of a first real number and the sum of a second and third real number *equals* the sum of the product of the first and second and the product of the first and third*].* □

The following theorem is a slight variation of the previous axiom.

Theorem 160. *Prove that for all x, y, and z, if x, y, and z are real numbers, then*

$$(x+y) \cdot z = (x \cdot z) + (y \cdot z)\,.$$

Proof. Assume that x, y, and z are real numbers. Then

$$\begin{array}{rcll} (x+y)\cdot z & = & z\cdot(x+y) & \text{by Axiom 154} \\ & = & (z\cdot x)+(z\cdot y) & \text{by Axiom 159} \\ & = & (x\cdot z)+(z\cdot y) & \text{by Axioms 154 and 156} \\ & = & (x\cdot z)+(y\cdot z) & \text{by Axioms 154 and 156} \end{array}$$

By (repeated uses of) Axiom 137, $(x+y)\cdot z = (x\cdot z)+(y\cdot z)$. □

Before we proceed to the next items, there are some style notes to address, regarding the proof of Theorem 160.

Much like the proof of Theorem 158, the proof of Theorem 160 features equals signs that have been aligned, which is, again, for the sake of neatness. However, in the proof of Theorem 160, only the top line appears with a left-hand side to the equation. This is because each newly written quantity equals the quantity written above it. Therefore, the computational-looking portion of the above proof could be read as "$(x+y)\cdot z$ equals $z\cdot(x+y)$ (by

Axiom 154), which equals $(z \cdot x) + (z \cdot y)$ (by Axiom 159), which equals $(x \cdot z) + (z \cdot y)$ (by Axioms 154 and 156), which equals $(x \cdot z) + (y \cdot z)$ (by Axioms 154 and 156)."

Overall, the format of the proof of Theorem 160 is useful in situations where we need to prove that an equation is true, but we don't have a hypothesis that is itself an equation. We instead showed that the left-hand side of our desired conclusion equals some quantity, which equals some other quantity, ..., which eventually equals the right-hand side of our desired conclusion (listing reasons along the way). Working behind the scenes in such instances is Axiom 137, which asserts that the first and last of the quantities in that narrative equal each other (by virture of being equal to every quantity along the way). This is another instance of a less critical axiom being cited, so feel free to remove the bottom line of the proof entirely ("By (repeated uses of) Axiom 137...").

Additionally, the references to Axiom 156 in the proof of Theorem 160 can be removed, as the use of Axiom 154 is more critical on those two lines. For example, in the third of the four lines that contain an equals sign, it's more interesting that the order of z and x has changed in the first product (versus the line above) than noticing that $z \cdot y$ has not changed. (It's as if $z \cdot y$ from the theorem is acting as z from Axiom 156.)

We continue with some issues related to the behavior of 0 and of -1.

Problem 161. *In this problem, you will show that each real number has only one additive inverse. You will do this by proving the following: for all real numbers x and y, if $x + y = 0$, then $y = -x$. To start this proof, assume that $x + y = 0$, use Axiom 156 to add $-x$ to both sides of that equation, then determine what else must happen.* □

Problem 162. *Use Problem 161 to prove that for all x, if x is a real number, then $x = -(-x)$.* □

Problem 163. *Use Problem 161 to quickly prove that $0 = -0$ (so 0 is its own additive inverse). [The statement to be proven is not a conditional, so your proof should not start with an assumption.]* □

Problem 164. *Prove that for all x, if x is a real number, then $0 \cdot x = 0$. [Hint: after your assumption, start by justifying why $0 \cdot x = (0 + 0) \cdot x$.]* □

Problem 165. *Prove that for all x, if x is a real number, then $(-1) \cdot x = -x$. [Hint: first try to demonstrate that $x + ((-1) \cdot x) = 0$, then apply Problem 161.]* □

Axiom 166. -1 *is an integer.* □

Problem 167. *Use Problem 165, Axiom 166, and one other Chapter 3 result to quickly prove that for all x, if x is an integer, then $-x$ is an integer.* □

Problem 168. *Prove that for all x and y, if x and y are real numbers, then* $x \cdot (-y) = (-x) \cdot y$. □

Problem 169. *Prove that for all x and y, if x and y are real numbers, then* $(-x) \cdot (-y) = x \cdot y$. □

Before we proceed into vocabulary and problems that are more representative of number theory, here is one final algebraic type of problem: an extension of the results from Axiom 159 and Theorem 160.

Problem 170. *Prove the familiar FOIL formula: for all w, x, y, and z, if w, x, y, and z are real numbers, then*

$$(w+x) \cdot (y+z) = (w \cdot y) + (w \cdot z) + (x \cdot y) + (x \cdot z).$$

□

3.2 Even and Odd Integers

With some algebraic preliminaries in place, we are now ready for a more formal discussion of a fundamental topic from number theory: the familiar concept of *even* and *odd* integers.

Axiom 171. *There is an integer* 2 *(read out loud as "two") such that* $2 = 1 + 1$. □

Definition 172. *An integer x is called* even *if and only if* $x = 2 \cdot k$ *for some integer k.* □

Definition 173. *An integer x is called* odd *if and only if* $x = (2 \cdot k) + 1$ *for some integer k.* □

Based on these definitions, if you are to prove that an integer is even, then you must show that that integer equals the product of 2 and some integer, and if you are to prove that an integer is odd, then you must show that that integer equals the sum of such a product (of 2 and some integer) and 1.

As a small bit of guidance on the next six problems, remember that a direct proof of a universal conditional begins by assuming the hypothesis (and I recommend that you use direct proofs on all six). Once you have written down an appropriate assumption (about x or y being even or being odd), you should immediately use a relevant definition (172 or 173) to write x and y in a specific way. Then look at the conclusion of the statement that you're trying to prove. When you're asked to prove that $x+y$ or $x \cdot y$ is even or odd, then use the ways that you wrote x and y to determine a way that $x+y$ or $x \cdot y$ looks (and you will use absolutely *any* relevant axiom or theorem from earlier in this chapter to do so). These steps that I have prescribed will not complete the entire proof, but they will cover a lot of the necessary ground.

Problem 174. *Prove that for all integers x and y, if x and y are even, then $x+y$ is even.* □

Problem 175. *Prove that for all integers x and y, if x is even and y is odd, then $x+y$ is odd.* □

Problem 176. *Prove that for all integers x and y, if x and y are odd, then $x+y$ is even.* □

Problem 177. *Prove that for all integers x and y, if x and y are even, then $x \cdot y$ is even.* □

Problem 178. *Prove that for all integers x and y, if x is even and y is odd, then $x \cdot y$ is even.* □

Problem 179. *Prove that for all integers x and y, if x and y are odd, then $x \cdot y$ is odd.* □

Note that for the next three problems, the statements to be proven are not conditionals, so your proofs should not start with an assumption.

Problem 180. *Prove that* 0 *is even.* □

Problem 181. *Prove that* 1 *is odd.* □

Problem 182. *Prove that* 2 *is even.* □

Problem 183. *Prove that for all integers x, if x is even, then $-x$ is even.* □

Problem 184. *Prove that for all integers x, if x is odd, then $-x$ is odd.* □

3.3 Squares, Cubes, and the Rest of the Integers

We begin this section with a few results relating to *squares* and *cubes*.

Definition 185. *Let x be a real number. The real number x^2, read out loud as "x squared," is defined to be the product of x and itself: $x^2 = x \cdot x$. The real number x^3, read out loud as "x cubed," is defined to be the product of x^2 and x: $x^3 = \left(x^2\right) \cdot x$.* □

Problem 186. *Prove that for all real numbers x, $(-x)^2 = x^2$.* □

Problem 187. *Prove that for all real numbers x, $(-x)^3 = -\left(x^3\right)$.* □

Problem 188. *Prove that for all integers x, if x is odd, then x^3 is odd.* □

Setting aside squares and cubes for the moment, we note that after having carefully developed the existence and behavior of the integers 1, 0, 2, and -1, the following axiom finally extends the discussion of integers into the collection of all known familiar "whole numbers."

Axiom 189. *The list*

$$\ldots,-10,-9,-8,-7,-6,-5,-4,-3,-2,-1,0,1,2,3,4,5,6,7,8,9,10,\ldots$$

constitutes all of the integers, where the ellipses at each end of the list indicate that the list continues without terminating. □

From our study in this chapter, we know that the sum of 1 and 2 is an integer, being the sum of two integers. The above axiom is giving that sum a name (the name 3). Similarly, the above axiom is giving the sum of 1 and 3 a name (the name 4), and so on for all of the integers further to the right in that list. Problem 167 then provides the names for all of the integers further to the left of 0 in the list. In all, Axiom 189 is a declaration that we will finally use our "whole numbers" to which we are accustomed, and in the ways that we are accustomed.

We close the chapter with a few problems that allow us to consider some specific integers. As all of these problems ask you to settle a conjecture, please re-visit Chapter 2 to remind yourself of what that means.

Problem 190. *Settle the conjecture "For all a, b, and c, if a, b, and c are integers, then* $a\cdot(b+c)=(a\cdot b)+c$*."* □

Problem 191. *Settle the conjecture "For all a and b, if a and b are integers, then* $(a+b)^2=a^2+b^2$*."* □

Problem 192. *Settle the conjecture "For all a, b, c, d, e, and f, if a, b, c, d, e, and f are integers, then* $(a+b)\cdot(c+d)\cdot(e+f)=(a\cdot c\cdot e)+(b\cdot d\cdot f)$*."* □

Chapter 4

Set Theory

In this chapter, we will discover formal proofs in the subject of set theory. Based on Definition 106 and Example 107, "set theory" is what we call the axiomatized study of objects known as "sets." Set theory is not only fundamental to any mathematician's training above the computational level, but it is also applicable in many computational circumstances, most notably in probability. If you have ever drawn a Venn diagram in your life (which I suspect you have), then you have a passing familiarity with some set theory. However, while Venn diagrams may be useful for helping us to consider conjectures at an intuitive level, we will make our reasoning formal through the kind of logical processes that have been mentioned in the earlier chapters.

4.1 Undefined Terms and Preliminary Examples of Sets

In order to prove results about sets, we must start with an axiom system. Recall from Definition 80 that an axiom system consists of undefined terms and axioms. We will take *set* and *element* as undefined terms. The treatment of set theory in this chapter will only use two axioms, both of which will be stated in later sections.

Warning: As has been addressed in the previous two chapters, mathematical conversations have to start with some terms that are undefined. However, both axioms that we will assume for sets and every result that we will derive from those axioms will agree with the interpretation that a set is a collection of concrete or abstract objects, and those objects are the elements of that set. With this interpretation in mind, we will agree that any collection of objects that we can intelligently describe will be regarded as a set. $\square$

Example 193. *The following all describe elements that belong to certain sets:*

1. *The number* 14 *is an element of the set of even numbers.*
2. *Hank Aaron is an element of the set of players in the Baseball Hall of Fame.*
3. *A baseball bat is an element of the set of TSA's prohibited carry-on items.* □

We use the notation $x \in A$ to say "x is an element of the set A." For example, if we let E be the set of even numbers, then we can write that $14 \in E$.

On the other hand, we write $x \notin A$ to say "x is not an element of the set A." (For example, $13 \notin E$.) Note that "$x \in A$" and "$x \notin A$" are each statements (negations of each other, at that).

4.2 Equality, Subset Containment

Our first means of relating one set to another is the following definition that addresses equality of sets.

Definition 194. *Let A and B be sets. We say that A* equals *B if and only if the statement "For all x, $x \in A$ if and only if $x \in B$" is true. The notation $A = B$ says "A equals B."* □

Here are a few immediate comments about this definition.

1. Since it can be shown that $P \leftrightarrow Q \equiv Q \leftrightarrow P$, applying that equivalence to this definition tells us that $A = B$ if and only if $B = A$. [If two sets are equal to each other, then the order in which we state the equality does not matter.]
2. As a consequence of this definition, a set is defined by what its elements are. [If you change the elements, then you've changed the set.]
3. We write $A \neq B$ to say "A and B are not equal to each other" or "A does not equal B." In Problem 203, you will carefully consider what this "inequality" means for sets.

When possible, it is customary to identify a set simply by listing the set's elements inside of braces, with the elements separated from each other by commas. For example, if we say that $A = \{1, 2, 3\}$, then we know the set A to be the set consisting only of the elements 1, 2, and 3. By Definition 194, we know that any other set that consists only of the elements 1, 2, and 3 is actually equal to the set that we've named A. As another example, if we let $B = \{\text{Wyoming}, \text{Idaho}\}$, then we know B to be the set consisting only of the elements Wyoming and Idaho.

Problem 195. *Let $C = \{1,2,3,4\}$ and $D = \{3,2,1,4\}$. Does C equal D? Briefly explain why or why not.* $\square$

Definition 196. *Let A and B be sets. We say that A is a* subset *of B if and only if every element of A is also an element of B. Saying that every element of A is also an element of B is the same as saying "For all x, if $x \in A$, then $x \in B$." The notation $A \subseteq B$ says "A is a subset of B."* $\square$

Example 197. *Let $R = \{1,2,3\}$ and $S = \{1,2,3,4,5,6\}$. We note that each element of R is also an element of S: $1 \in R$ and also $1 \in S$; $2 \in R$ and also $2 \in S$; $3 \in R$ and also $3 \in S$. Since those were all of the elements of R, we have (by Defintion 196) that $R \subseteq S$.* $\square$

Example 198. *Let E be the set of even numbers (from the previous chapter, these are numbers that can be written as $2 \cdot k$ for some integer k). Let T be the set of multiples of 10 (so that elements of T can be written as $10 \cdot k$ for some integer k). Let's address some of the main ideas of the proof of the following theorem: $T \subseteq E$. By Definition 196, we'd have to show that for all x, if $x \in T$, then $x \in E$. Note that this is a universal conditional. Noting the discussion between Definition 101 and Example 102, we get to assume for free that $x \in T$, but we must show that $x \in E$, as a consequence. Because we have assumed that $x \in T$, we have (by the definition of the set T) that $x = 10 \cdot k$ for some integer k. For the sake of brevity, I won't list the individual reasons, but the axioms of the previous chapter tell us that*

$$x = 10 \cdot k = (2 \cdot 5) \cdot k = 2 \cdot (5 \cdot k)$$

and that $5 \cdot k$ is an integer. By the definition of E, we now have that $x \in E$. What we have shown is that for all x, if $x \in T$, then $x \in E$. By Definition 196, T is a subset of E: $T \subseteq E$. $\square$

Problem 199. *Let's say that you have two completely arbitrary sets A and B (you know nothing up front about the elements of A or B; you simply know that A and B are a couple of sets that are named A and B). What is the simplest specific requirement on these sets that would guarantee that A is not a subset of B? To answer this question, first write down the first statement that is in quotation marks in Definition 196, and use the paragraph between Problems 70 and 71 to determine that statement's negation.* $\square$

Problem 200. *Using the sets R and S from Example 197, is S a subset of R? Why or why not?* $\square$

Problem 201. *Using the sets T and E from Example 198, is E a subset of T? Why or why not?* $\square$

Problem 202. *Fill in the blanks of the proof of the theorem "For all A, if A is a set, then $A \subseteq A$." [This theorem says that every set is a subset of itself.]*

Proof. Assume that A is a set. [We want to show that $A \subseteq A$. By Definition 196, this means that we need to show that for all x,
<u>if $x \in$ (blank), then $x \in$ (blank).</u>] This underlined statement is indeed true for every x, by Problem (blank) from Chapter 1. Therefore, by Definition 196, $A \subseteq A$. □

Problem 203. *Say that you have two arbitrary sets A and B. What is the simplest specific requirement on these sets that would guarantee that A and B are not equal? To answer this question, first write down the first statement that is in quotation marks in Definition 194, and use the paragraph between Problems 70 and 71 (in addition to a Chapter 1 problem about biconditionals, which you should find and cite) to determine that statement's negation.* □

Problem 204. *Take the conjecture "The set of NBA players is equal to the set of people who are over seven feet tall." Identify every type of person whose existence would demonstrate that this conjecture is false. [I'm not asking you to identify anyone by name, but instead by characteristics].* □

Problem 205. *Invent a set A. Invent a set B such that $A \subseteq B$. Invent a set C such that $B \subseteq C$. Looking only at your sets A and C (and ignoring your set B), do you have that $A \subseteq C$? Briefly explain why or why not, using only Defintion 196.* □

Problem 206. *Settle the following conjecture: for all sets A, B, and C, if $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$. [Because of the use of the word "all," Problem 205 will not supply the proof here.]* □

We now state a theorem that amounts to a very useful rephrasing of Definition 194.

Theorem 207. *Two sets are equal to each other if and only if each is a subset of the other: For all sets A and B, $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$.*

Proof. Assume that A and B are sets. By Definition 194, $A = B$ if and only if the statement "For all x, $x \in A$ if and only if $x \in B$" is true. By Definition 58, this statement is equivalent to "For all x, if $x \in A$, then $x \in B$, and if $x \in B$, then $x \in A$." By Definition 196, that statement is equivalent to "$A \subseteq B$ and $B \subseteq A$." □

Problem 208. *Prove that for all sets A, B, and C, if $A \subseteq B$, $B \subseteq C$, and $C \subseteq A$, then $A = B$, $B = C$, and $C = A$. [After stating your assumptions, apply Problem 206 and Theorem 207 in order to show that $A = C$. Then use those same two results to show that $A = B$. Repeat that process again to show that $B = C$.]* □

4.3 Subsets and Variable Statements Revisited

In Section 4.1 and Section 4.2, we covered some general results related to some fundamental ideas in set theory: elements, equality, and subsets. We take a slight step back in this section to look at some formalized means of describing the elements of particular sets.

As we saw just before Definition 196, when possible, we identify a set simply by listing the set's elements inside of braces, with the elements separated from each other by commas. For example, if we let $A = \{1, 2, 3\}$, then we know the set A to be the set consisting only of the elements 1, 2, and 3.

In other circumstances, it is impractical, if not impossible, to identify a set by listing its elements. For example, some sets (like the set of even numbers) have infinitely many elements, so any attempt to write down every element would be futile. As another example, we may want to name and discuss sets when we don't even know what their specific elements are. In an attempt to address these two situations, we introduce the following axiom.

Axiom 209. *For any set A and any variable statement $P(x)$, there is a set B whose elements are exactly those elements x of A for which the statement $P(x)$ is true.* □

At its heart, Axiom 209 allows us to select a subset B of the set A based upon the truth of a variable statement $P(x)$.

Using the names from Axiom 209, we notate B by writing $B = \{x \text{ in } A \,|\, P(x)\}$. Let's identify the meaning of each piece of this notation. As always, the braces indicate that we're talking about a set, so B is a set. We read the vertical bar as the phrase "such that." When we write an equality $B = \{x \text{ in } A \,|\, P(x)\}$, we are saying, "$B$ is the set of all elements x in the set A such that $P(x)$ is a true statement."

Problem 210. *Let $P(x)$ be the variable statement "x has more than one million residents." Let A be the set of all U.S. cities. List the elements of the set B, where $B = \{x \text{ in } A \,|\, P(x)\}$. I will allow online fact-checking for this problem.* □

The notation from Axiom 209 allows us to revisit the discussion of real numbers, as they relate to *interval* notation, a topic that you may recall from previous math courses. First note that it is common to notate the set of all real numbers as $\mathbb{R}$. It is important to recognize that $\mathbb{R}$ is the name of a special set, and not to be used as a casual abbreviation for the word "real." It is a common error to say "the set of all $\mathbb{R}$ numbers," when "$\mathbb{R}$" alone is what is meant. Given this name for the set of all real numbers, we can define intervals as special subsets of $\mathbb{R}$.

Definition 211. *Let a and b be real numbers, where $a < b$. An* interval *is a subset of the real number line that has any of the following eight forms:* (a,b)*,* $[a,b)$*,* $(a,b]$*,* $[a,b]$*,* (a,∞)*,* $[a,\infty)$*,* $(-\infty,b)$*, or* $(-\infty,b]$*, where*

$$\begin{array}{rrcl} & (a,b) & = & \{x \text{ in } \mathbb{R} \,|\, a < x < b\}, \\ & [a,b) & = & \{x \text{ in } \mathbb{R} \,|\, a \leq x < b\}, \\ & (a,b] & = & \{x \text{ in } \mathbb{R} \,|\, a < x \leq b\}, \\ & [a,b] & = & \{x \text{ in } \mathbb{R} \,|\, a \leq x \leq b\}, \\ & (a,\infty) & = & \{x \text{ in } \mathbb{R} \,|\, a < x\}, \\ & [a,\infty) & = & \{x \text{ in } \mathbb{R} \,|\, a \leq x\}, \\ & (-\infty,b) & = & \{x \text{ in } \mathbb{R} \,|\, x < b\}, \\ \text{and} & (-\infty,b] & = & \{x \text{ in } \mathbb{R} \,|\, x \leq b\}. \end{array}$$

□

For example, the interval $(1,3]$ is the set of all elements x in $\mathbb{R}$ such that $1 < x \leq 3$. As a quick note, we have that $3 \in (1,3]$, but $1 \notin (1,3]$. We will use interval notation for various problems in the remainder of Chapter 4.

Before we proceed to other topics, though, we will note a related subset containment. It is common to notate the set of all integers as $\mathbb{Z}$. Putting Definition 196 together with Axiom 133, we then see that $\mathbb{Z}$ is a subset of $\mathbb{R}$.

The following definition addresses a particularly special set.

Definition 212. *Let A be a set. Let $P(x)$ be the variable statement "x is not x," which is false, no matter which x you consider. By Axiom 209, there is a subset $\{x \text{ in } A \,|\, x \text{ is not } x\}$ of A, and it would contain no elements at all. Call this set the* empty set*, which is notated either as the symbol $\varnothing$ or as a stand-alone pair of curly braces $\{\,\}$.* □

Warning: The symbol $\varnothing$ is the lower-case Greek letter phi, not the number zero. □

Definition 194 indicates that any two sets without elements are actually the same set. Therefore, there is only one empty set, which is why we can get away with saying "*the* empty set." For example, the set of 8-term U.S. Presidents is the same as the set of 1000-time Academy Award winners, since both of those two sets are actually the empty set.

The empty set $\varnothing$ presents some interesting (that is, odd) situations.

Example 213. *Let P be the absurd conjecture "Every real-life unicorn has served on the Supreme Court." We can rephrase P as "The set U consisting of all the real-life unicorns is a subset of S, which is the set of all current and former Supreme Court Justices." Using Definition 196, we can rephrase P yet again as the universal conditional "For all x, if $x \in U$, then $x \in S$." Because the set U is empty, the statement "$x \in U$" is false for all x. Therefore, the conditional "If $x \in U$, then $x \in S$" is* true *for all x, because the hypothesis is false for all x, which proves that U is a subset of S. The absurd conjecture P is actually true!* □

Problem 214. *Using Example 213 as a guide if necessary, prove that the empty set is a subset of any set. That is, you are to prove that for all A, if A is a set, then $\varnothing \subseteq A$.* □

Problem 215. *Show that for all A, if $A \subseteq \varnothing$, then $A = \varnothing$. This says that any subset of the empty set is the empty set itself. [The fastest proof of this theorem will use the result of the previous problem, along with a theorem from earlier in Chapter 4, so that you won't have to write "x" at all].* □

Problem 216. *Briefly explain the difference between the set $\varnothing$ and the set $\{\varnothing\}$ by saying what each set's elements are.* □

4.4 Unions and Intersections

In this section, we will consider methods of using existing sets to create more complicated sets. This should remind you of our work in Chapter 1, where we built complicated statements out of simpler statements. Back in that chapter, we performed such constructions by using the simple words "and," "or," and "not" (among others). Each of these three little words from Chapter 1 will correspond to a set theory topic in the remainder of Chapter 4.

We get that discussion moving by taking the following axiom.

Axiom 217. *For any two sets, there is a set that contains both as subsets.* □

This axiom says that if we have two sets A and B, then there is a set $\mathscr{U}$ such that $A \subseteq \mathscr{U}$ and $B \subseteq \mathscr{U}$.

Definition 218. *Let A and B be sets. Axiom 217 guarantees the existence of a set $\mathscr{U}$ that satisfies $A \subseteq \mathscr{U}$ and $B \subseteq \mathscr{U}$. Call such a set $\mathscr{U}$ a* parent set *to A and B.* □

Definition 219. *Let A and B be sets, and let $\mathscr{U}$ be a parent set. By Axiom 209, there exists a set*

$$\{x \text{ in } \mathscr{U} \,|\, x \in A \text{ or } x \in B\}.$$

Call this set the union *of A and B. We notate the union of A and B as $A \cup B$, and we often omit the parent set (Axiom 217 reminds us that it's there, but you can see in Problem 265 why we can omit it):*

$$A \cup B = \{x \,|\, x \in A \text{ or } x \in B\}.$$

We can alternately state (in the language of logic), "For all x, $x \in A \cup B$ if and only if $x \in A$ or $x \in B$." With disjunctions (statements using the word "or") in mind, an element x in $A \cup B$ must be an element of at least one of the sets A and B. □

Definition 220. *Let A and B be sets, and let $\mathscr{U}$ be a parent set. By Axiom 209, there exists a set*

$$\{x \text{ in } \mathscr{U} \mid x \in A \text{ and } x \in B\}.$$

Call this set the intersection *of A and B. We notate the intersection of A and B as $A \cap B$, and we often omit the parent set (as we did in the previous definition):*

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}.$$

We can alternately state, in the language of logic, that for all x, $x \in A \cap B$ if and only if $x \in A$ and $x \in B$. Therefore, an element x in $A \cap B$ must simultaneously be an element of A and of B. □

To help you remember how to tell the symbols $\cup$ and $\cap$ apart, notice that $\cup$ looks like the letter U, which stands for "union." Also, note that the union and intersection symbols $\cup$ and $\cap$ are simply rounded versions of the disjunction ("or") and conjunction ("and") symbols $\vee$ and $\wedge$, respectively. This is not a coincidence, as unions are defined in terms of disjunctions and intersections are defined in terms of conjunctions.

Problem 221. *Write each of the following in words (not with symbols, but exactly as each of these would be said out loud) so that you can determine which are statements and which aren't statements. Briefly explain your choices, using complete sentences.*

1. $A = B$
2. $A \cap B$
3. $A \cup B$
4. $A \subseteq B$ □

Warning: Be advised that none of our proofs in this section will require any mention of a parent set. We only mentioned a parent set because Axiom 217 allows us to see that unions and intersections of sets exist and can be described. □

Problem 222. *Let $A = \{a,e,i,o,u\}$ and $B = \{a,b,c,d,e,f,g\}$. Find $A \cup B$ and $A \cap B$. [You should notate these answers by listing the correct elements inside of braces, separated from each other by commas.]* □

Problem 223. *Consider the intervals $(0,5]$ and $[1,6)$. Find their union and find their intersection, perhaps by drawing a diagram. [You should notate these answers by using interval notation, not by listing the correct elements inside of braces, separated from each other by commas.]* □

Theorem 224. *For all A and B, if A and B are sets, then $A \cup B = B \cup A$.*

I will first provide a proof that contains extra explanations.

Proof. As with any theorem, we assume that the hypothesis is true, so assume that A and B are sets. By Definition 219,

$$A \cup B = \{x \,|\, x \in A \text{ or } x \in B\}.$$

By Part 2 of Problem 31, we know that the statement "$x \in A$ or $x \in B$" is equivalent to the statement "$x \in B$ or $x \in A$," so the values of x that make "$x \in A$ or $x \in B$" true are the same values of x that make "$x \in B$ or $x \in A$" true. Therefore,

$$\begin{aligned} A \cup B &= \{x \,|\, x \in A \text{ or } x \in B\} \\ &= \{x \,|\, x \in B \text{ or } x \in A\} \\ &= B \cup A, \end{aligned}$$

where, for the last equality, we simply used Definition 219 again (this time, with the names A and B swapped). □

I will now provide a more concise proof of Theorem 224. This amount of writing will be appropriate when you write a proof that two differently written sets are, in fact, equal to each other.

Proof. Assume that A and B are sets. Then

$$\begin{array}{rcll} A \cup B & = & \{x \,|\, x \in A \text{ or } x \in B\} & \text{by Definition 219} \\ & = & \{x \,|\, x \in B \text{ or } x \in A\} & \text{by Part 2 of Problem 31} \\ & = & B \cup A & \text{by Definition 219} \end{array}$$

□

This more consise proof would be read out loud from start to finish as follows: "Assume that A and B are sets. Then the union of A and B equals the set of all x such that x is an element of A or x is an element of B (by Definition 219), which equals the set of all x such that x is an element of B or x is an element of A (by Part 2 of Problem 31), which equals the union of B and A (by Definition 219)."

Theorem 224 essentially boils down to the idea that the two statements that identify the elements of the sets $A \cup B$ and $B \cup A$ are actually equivalent statements. Notice, in fact, that the claim made in Theorem 224 ($A \cup B = B \cup A$) looks very similar to the claim made in Part 2 of Problem 31 ($P \vee Q \equiv Q \vee P$).

Theme: Because unions and intersections are defined in terms of the logic words "or" and "and," we will notice that many proofs of results about sets will use facts from Chapter 1, which is where we manipulated logical operations. It will often be the case that an equivalence of statements leads to an equality of sets (which is what happened from the top line of centered equalities in the more concise proof of Theorem 224 down to the middle line). If you have been asked to prove an equality that contains $\cup$ or $\cap$ symbols, then you should look for a similarly stated equivalence in Chapter 1 (but with $\vee$ or $\wedge$ symbols instead). $\square$

Problem 225. *Prove (using the format of the concise version of the proof of Theorem 224) that for all A and B, if A and B are sets, then* $A \cap B = B \cap A$. $\square$

Problem 226. *Invent three sets A, B, and C, such that* $A \cup B = C$ *and* $A \cap B = \varnothing$. $\square$

Problem 227. *For the sets A and B in Problem 222, is it true that* $A \subseteq (A \cup B)$*? Briefly explain why, using only Definition 196. Make sure to list the elements of the relevant sets again.* $\square$

The following theorem generalizes the previous result.

Theorem 228. *For all A and B, if A and B are sets, then* $A \subseteq (A \cup B)$.

As with Theorem 224, I will first provide a proof that contains extra explanations.

Proof. As we do when we prove a universal conditional, assume the hypothesis: assume that A and B are sets. We are trying to show that $A \subseteq (A \cup B)$, which (by Definition 196) means that we are trying to show that for all x, if $x \in A$, then $x \in (A \cup B)$. Since this is itself a universal conditional, let's assume the hypothesis: assume that $x \in A$. In this case, the disjunction "$x \in A$ or $x \in B$" is true because at least the component "$x \in A$" is true. Then by Definition 219, we have arrived at the desired conclusion that $x \in (A \cup B)$, since the statement "$x \in A$ or $x \in B$" has to be true in order for the statement "$x \in (A \cup B)$" to be true. By Definition 196, $A \subseteq (A \cup B)$. $\square$

I will now provide a more concise proof of Theorem 228. This amount of writing will be appropriate when you write a proof that one set is a subset of another.

Proof. Assume that A and B are sets. Assume that $x \in A$. Then $x \in A$ or $x \in B$. By Definition 219, $x \in (A \cup B)$. By Definition 196, $A \subseteq (A \cup B)$. $\square$

Theme: When you are trying to prove that one set is a subset of a second set, you are free to use this same technique of assuming that you have an element in the first set. Then use any other facts at your disposal to show that your arbitrary element from the first set has to be in the second set as well. $\square$

Problem 229. *Prove (using the format of the concise version of the proof of Theorem 228) that for all A and B, if A and B are sets, then* $(A \cap B) \subseteq A$. ☐

Similarly, $(A \cap B) \subseteq B$, but I am not asking you to prove this.

Problem 230. *Prove that for all A and B, if A and B are sets, then* $(A \cap B) \subseteq (A \cup B)$. *Instead of using the technique from the theme above, use only Theorem 228, Problem 229, and one problem from Section 4.2 in your proof.* ☐

Problem 231. *If we know nothing about the elements of a set A, can we still say something special about the set* $A \cup \varnothing$? *Try finding this union for some specific set A, but then prove that you must get the same results no matter which A you could possibly use.* ☐

Problem 232. *Repeat the previous problem, but investigate* $A \cap \varnothing$ *instead.* ☐

Problem 233. *Settle the conjecture: for all sets A and B, if* $A \cup B = A$, *then* $A = B$. ☐

Problem 234. *Identify a property that two arbitrary sets A and B could satisfy that would guarantee that* $A \cup B = A$. *Phrased another way, fill in the blank of the theorem "For all A and B, if (blank), then* $A \cup B = A$," *and prove that theorem. [To figure out how to fill in the blank, I recommend that you invent some sets A and B, where* $A \cup B = A$, *and try this several times, which you likely did for the previous problem. Try to find some sort of common pattern among all of your examples.]* ☐

Problem 235. *Recall from Example 198 the set E of all even numbers and the set T of all multiples of* 10. *Find* $T \cap E$. *Briefly explain your reasoning. [Note that* $T \cap E$ *will actually be a set whose name you already know.]* ☐

Problem 236. *Prove that for all sets A and B, if* $A \subseteq B$, *then* $A \cap B = A$. ☐

Problem 237. *Prove that for all sets A and B, if* $A \cap B = A$, *then* $A \subseteq B$. ☐

Problem 238. *Prove that for all sets A, B, and C, if* $A \subseteq C$ *and* $B \subseteq C$, *then* $(A \cup B) \subseteq C$. ☐

Problem 239. *Invent three sets A, B, and C such that none of those sets is a subset of either of the other two, and every two of those three sets share at least one element in common, but there is no element that is common to all three of the sets. Find each of the sets* $B \cup C$, $A \cap B$, $A \cap C$, $A \cap (B \cup C)$ *and* $(A \cap B) \cup (A \cap C)$. *What do you notice about these last two sets?* ☐

Problem 240. *Settle the conjecture: for all A, B, and C, if A, B, and C are sets, then* $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$. *[Re-read the theme between Theorem 224 and Problem 225 first.]* ☐

Problem 241. *Devise and settle a similar conjecture for $A \cup (B \cap C)$, for all sets A, B, and C.* □

Problem 242. *Assume that A and B are arbitrary sets such that $A \cup B = A \cap B$. What must consequently be true about A and B? That is, formulate a conjecture about A and B, and settle that conjecture.* □

4.5 Complements

We end our set theory content on a seemingly sad topic. The *complement* of a set is math's way of talking about "what's been left out."

Definition 243. *Let A and U be sets such that $A \subseteq U$. We define the* complement *of the set A (relative to U) to be the set*

$$A^c = \{x \text{ in } U \mid x \notin A\},$$

that is, the set of all elements of U that aren't elements of A. □

By this definition, we have that $A^c \subseteq U$.

Example 244. *Let U be the set of U.S. Presidents, and let A be the set of U.S. Presidents who served at least one full term, so that $A \subseteq U$. We can say that William Henry Harrison is an element of A^c because he fell ill at his inauguration and died approximately one month later, thereby failing to complete one full term.* □

Problem 245. *Let $U = \{1,2,3,4,5,6,7,8,9\}$, and let $A = \{1,3,5,7,9\}$, so that $A \subseteq U$. Find A^c.* □

Problem 246. *Using A and U from the previous problem, find $A \cap (A^c)$.* □

Problem 247. *Find the complement of the interval $[4,7)$, relative to $\mathbb{R}$, perhaps by drawing a diagram. [You should notate this answer by using interval notation.]* □

Problem 248. *Find the complement of $(-1,5] \cup (10,\infty)$, relative to $\mathbb{R}$, perhaps by drawing a diagram. [You should notate this answer by using interval notation.]* □

For the remaining problems in this section, assume that all complements are taken relative to U, where all we know about U is that it is a set and that it is named "U."

Problem 249. *Settle the conjecture: for all sets A, if $A \subseteq U$, then $A \cap (A^c) = \varnothing$.* □

Problem 250. *For all sets A that are a subset of U, what do you think that $A \cup (A^c)$ should always equal? Prove your conjecture.* □

Problem 251. *Prove that for all sets A, if $A \subseteq U$, then $(A^c)^c = A$.* $\square$

Problem 252. *What is the complement of U, relative to U? Prove your conjecture.* $\square$

Problem 253. *Prove that for all sets A and B, if $A \subseteq B$, then $(B^c) \subseteq (A^c)$.* $\square$

Problem 254. *As a variation of the previous problem, prove the related result: for all sets A and B, if $(B^c) \subseteq (A^c)$, then $A \subseteq B$.* $\square$

Problem 255. *Prove (using the format of the concise version of the proof of Theorem 224) that for all sets A and B, if A and B are each subsets of some set U, then $(A \cap B)^c = (A^c) \cup (B^c)$.* $\square$

Problem 256. *Devise and prove a similar result for $(A \cup B)^c$.* $\square$

Problem 257. *Settle the conjecture: for all sets A and B, if $A \cap B$ is the empty set, then $(A^c) \cap (B^c)$ is the empty set.* $\square$

4.6 Problems

Here are some spare problems to close out the chapter.

Problem 258. *Identify or create a set whose elements are sets themselves.* $\square$

Problem 259. *Explain why $T \notin E$, where T and E are described in Example 198.* $\square$

Problem 260. *Let's say that I attempt to prove the theorem from Example 198 by saying that $10 \in T$ and $10 \in E$, $20 \in T$ and $20 \in E$, $30 \in T$ and $30 \in E$, etc. What's wrong with this approach? Think about a major difference between the sets T and E versus sets like R and S from Example 197.* $\square$

Problem 261. *Invent two sets A and B (that is, list their elements), where $A \in B$ and $A \subseteq B$.* $\square$

Problem 262. *Look at the axiom system described in Example 102. Replace every instance of the word "Team" with the word "set" and every instance of the phrase "is on" with the phrase "is an element of." With the prescribed replacements, write the new version of each of the axioms given in that problem in terms of subset containment. That is, use the word "subset" or the $\subseteq$ symbol when appropriate.* $\square$

Problem 263. *Using the replacements prescribed in the previous problem, write and prove the new version of the theorem given in Example 102, using the vocabulary/notation of set theory.* $\square$

Problem 264. *Given any two sets A and B, Axiom 217 lets us assert the existence of a set $\mathscr{U}$ that contains A and B as subsets. Prove that given any three sets A, B, and C, there is a set $\mathscr{U}$ that contains all three as subsets. You may have to call upon Axiom 217 more than once.* $\square$

Problem 265. *Prove that for all sets A and B, if $\mathscr{U}$ and $\mathscr{V}$ are both parent sets to both of A and B, then*
$\{x \text{ in } \mathscr{U} \mid x \in A \text{ or } x \in B\} = \{x \text{ in } \mathscr{V} \mid x \in A \text{ or } x \in B\}$. $\square$

Problem 266. *State and settle a conjecture regarding an equality of sets that has not been mentioned in these notes.* $\square$

4.7 Chapter Summary

We used the following axiom system for set theory.

The terms *set* and *element* were undefined.

We assumed the following:

1. Axiom 209: For any set A and any variable statement $P(x)$, there is a set B whose elements are exactly those elements x of A for which the statement $P(x)$ is true.
2. Axiom 217: For any two sets, there is a set that contains both as subsets.

We could devise and state more axioms in order to prove more results. Also, we could certainly prove more results using only the axioms above. We will decelerate at this point, though, and recount some of the major results that we have seen:

For all sets A, B, and C, the following eight statements are true (where the complements are taken relative to some appropriate set):

1. $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$.
2. If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.
3. $\varnothing \subseteq A$ for any A.
4. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
5. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
6. $A \subseteq B$ if and only if $(B^c) \subseteq (A^c)$.
7. $(A \cap B)^c = (A^c) \cup (B^c)$
8. $(A \cup B)^c = (A^c) \cap (B^c)$

We close by noting some connections between Chapter 4 and Chapter 1.

set concept	related logic concept	connection
subset containment	conditional	Saying that $A \subseteq B$ is the same as saying that for all x, if $x \in A$ then $x \in B$.
set equality	biconditional	Saying that $A = B$ is the same as saying that for all x, $x \in A$ if and only if $x \in B$.
union	disjunction	Saying that $x \in A \cup B$ is the same as saying that $x \in A$ or $x \in B$.
intersection	conjunction	Saying that $x \in A \cap B$ is the same as saying that $x \in A$ and $x \in B$.
complement	negation	Saying that $x \in A^c$ is the same as saying that x is not in A.